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12. HAZARD TO LIFE

12.1 Introduction

12.1.1.1 This section of the EIA presents the analysis and findings of the hazard to life assessment undertaken for the Project.

12.1.1.2 In accordance with Section 3.5.12 of the EIA Study Brief (ESB-373/2025), a hazard to life assessment should be conducted to evaluate the risks associated with the use, transport and storage of explosives during construction of the Project.

12.1.1.3 Drill-and-blast excavation method will be adopted for the construction of tunnels for this Project. In addition, a magazine, to be constructed by the MTRC for Northern Link Project, will be handed over for the use of this Project to secure the explosive supply, allowing flexible blasting time and multiple faces under different excavation sequence to give a maximum tunnel production rate. The hazard to life assessment presented in this section addressed the following:

- Overnight storage of explosives in temporary magazine;
- Transport of explosives from the magazine to the explosives unloading points;
- Transport of explosives from explosives unloading points to the blast faces; and
- Use of explosives during the construction of the tunnels.

12.2 Legislation Requirements and Evaluation Criteria

12.2.1.1 The key legislation and guidelines that are considered relevant to the hazard to life assessment for the Project are as follows:

- Technical Memorandum on Environmental Impact Assessment Process (EIAO-TM);
- Dangerous Goods Ordinance (Cap. 295); and
- The EIA Study Brief (ESB-373/2025) Section 3.5.12 and Appendix K.

12.2.2 Technical Memorandum on Environmental Impact Assessment Process (EIAO-TM)

12.2.2.1 Annex 4 of the EIAO-TM specifies the individual and societal risk guidelines.

12.2.2.2 Individual risk (IR) is the predicted increase in the chance of fatality per year to an individual due to a potential hazard. The individual risk guidelines require that the maximum level of off-site IR should not exceed 1 in 100,000 per year, i.e. 1×10^{-5} per year.

12.2.2.3 Societal risk expresses the risks to the whole population. The Hong Kong Risk Guidelines (HKRG) is presented graphically in Diagram 12-1. It is expressed in terms of lines plotting the frequency (F) of N or more deaths in the off-site population from incidents at the installation. Two F-N risk lines are used in the HKRG that

demark “acceptable” or “unacceptable” societal risks. The intermediate region indicates the acceptability of societal risk is borderline and should be reduced to a level which is “as low as reasonably practicable” (ALARP). It seeks to ensure that all practicable and cost-effective measures that can reduce risk will be considered.

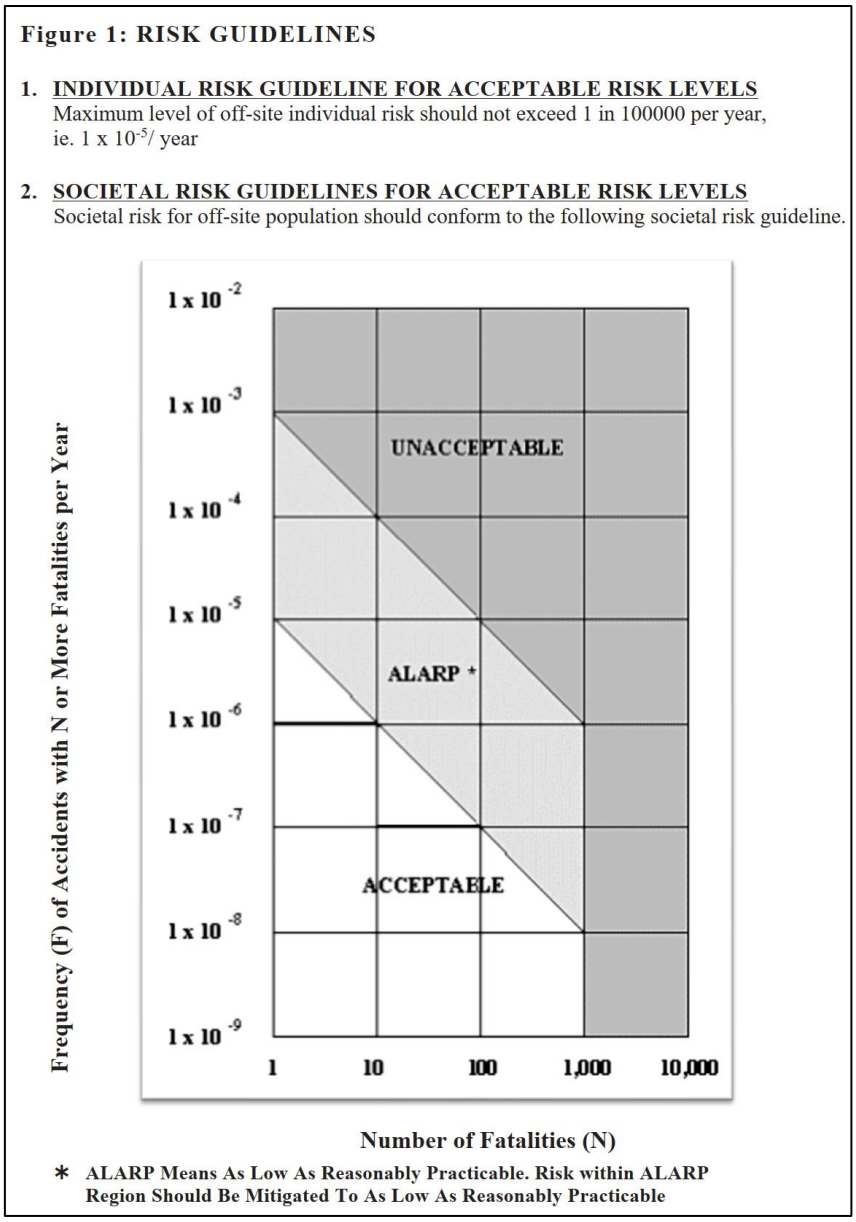


Diagram 12-1 Hong Kong Risk Guidelines

12.2.3 Dangerous Goods Ordinance

12.2.3.1 Storage of explosives is governed by the Dangerous Goods (General) Regulations (Cap. 295G). Under the regulation, a license is required for the storage of Class 1 dangerous goods (Class 1 S1DG).

12.2.3.2 The conveyance of explosives in Hong Kong is governed by the Dangerous Goods (Control) Regulation (Cap. 295G). A conveyance permit is required for any land transport of Class 1 dangerous goods (Class 1 S1DG) on public roads.

12.2.3.3 The explosives delivery vehicle should be an approved type complying with the regulations stated in the "Guidance Note No. GN2 Approval of an Explosives Delivery Vehicle" published by Mines Division.

12.2.3.4 The use of explosives in Hong Kong is also governed by the Dangerous Goods (Control) Regulation (Cap. 295G). A use (individual) license is required for an individual to use S1DG and a blasting permission is required in relation to a blasting operation in which S1DG are used.

12.2.4 EIA Study Brief (ESB-373/2025)

12.2.4.1 The detailed requirements of the hazard to life assessment are given in Section 3.5.12 and Appendix K of the EIA study brief. The main requirements are:

- Identify hazardous scenarios associated with the use, transport and storage of explosives during construction of the Project with a view to determine a set of relevant scenarios to be included in a Quantitative Risk Assessment (QRA);
- Execute a QRA of the set of hazardous scenarios determined, expressing population risks in both individual and societal terms;
- Compare individual and societal risks with the criteria for evaluating hazard to life stipulated in Annex 4 of the TM; and
- Identify and assess practicable and cost-effective risk mitigation measures to demonstrate the compliance with the risk guidelines stipulated in Figure 1 of Annex 4 of the TM.

12.3 Project Description

12.3.1 Overview

12.3.1.1 The construction of the Project is tentatively scheduled to be commenced in 2027 for completion in 2036. Blasting related construction activities include the construction of four road tunnels (**Diagram 12-2**). The use of explosives is expected to be between 2029 and 2033. Assessment year for construction phase is taken as 2033.

12.3.1.2 Explosives will be transported directly from the Kau Shat Wan Government Explosives Depot to the magazine site in Tai Shu Ha on a daily basis by the Mines Division. Staffs of Mines Division will unload the explosives from their explosives delivery vehicles and hand over the explosives to the contractor at the magazine. The contractor will be responsible for loading the explosives onto their explosives delivery vehicles at the magazine, transporting the explosives on public roads to three explosives unloading points, and then distributing to various blasting sites.

12.3.1.3 No explosives will be required during the operation phase of the Project, and thus risk assessment associated with the storage, transport and use of explosives is not required for operation phase of the Project.

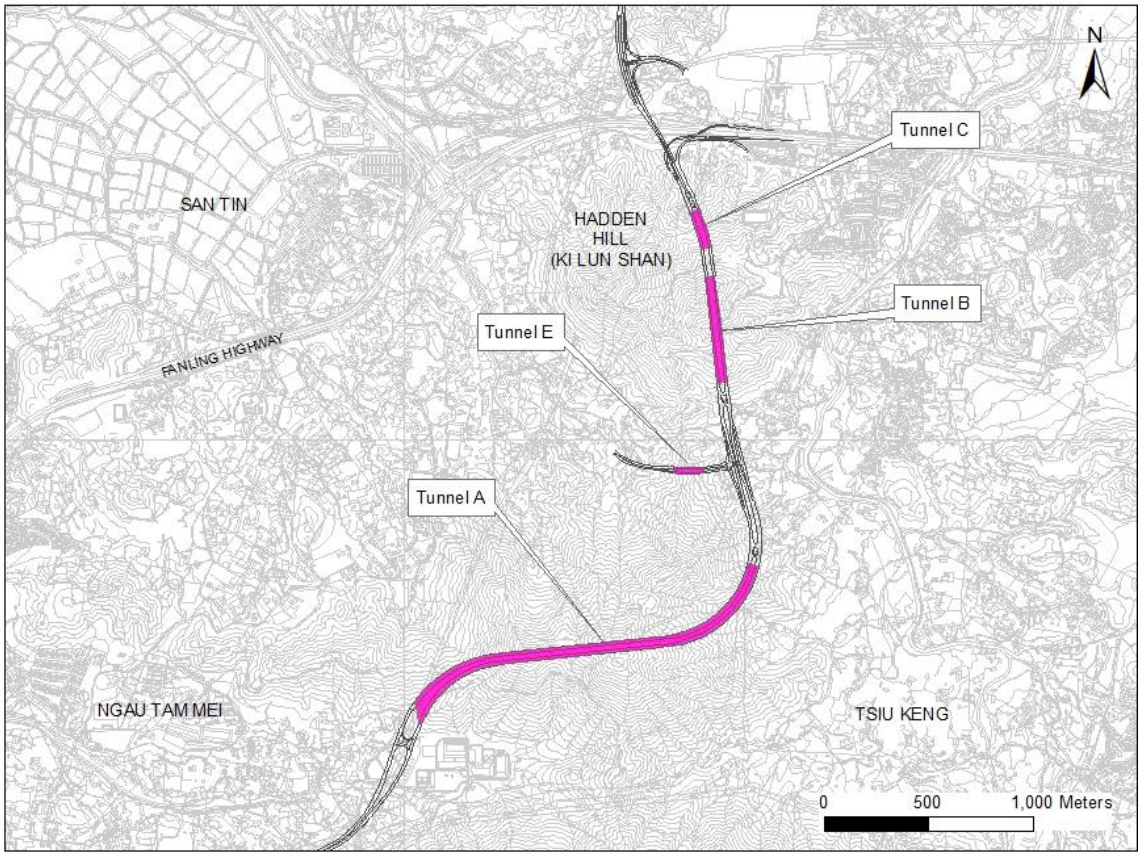


Diagram 12-2 Locations of Tunnels

12.3.2 Blasting Requirement

12.3.2.1 The tunnel construction will follow 1 blast per day for each blast face from both portals of a tunnel.

12.3.2.2 Tunnels A, B and C are double tube with 3-land carriageways for each bound; while Tunnel E is double tube with 2-land carriageways for each bound.

12.3.2.3 The blasting activities together with the required amount of explosives are summarised in **Table 12.1**. The actual amount of explosives is calculated based on different tunnel profiles described in **Table 12.2**.

Table 12.1 Blasting Requirement

Works Area	Blast Face	Length of Tunnel to be Blasted (km)	Approximate No. of Blasts	Explosives Load (kg/blast)
Tunnel A	3-lane	2.00	1710	49.4-1448.9
Tunnel B	3-lane	0.55	760	49.4-1448.9
Tunnel C	3-lane	0.20	100	49.4-1448.9
Tunnel E	2-lane	0.20	120	40.1-768.6

Table 12.2 Typical Tunnel Profiles and Explosives Requirement

Tunnel Type	Section Area (m^2)	No. of Production Holes	No. of Perimeter Holes	Cartridged Emulsion (TNT eqv. kg)	Bulk Emulsion (TNT eqv. kg)	Detonating Cord (TNT eqv. kg)	Detonators (TNT eqv. kg)
3-lane (CE)	160-200	191-233	64-73	407.4-491.5	N/A	14.4-16.4	0.72-0.86
3-lane (BE)	160-200	145-215	64-73	9.6	1154.8 – 1393.9	39.5-45.0	0.30-0.41
2-lane (CE)	122	124	49	258.4	N/A	10.98	0.48
2-lane (BE)	122	115	49	9.6	728.5	30.2	0.23

Notes:
1. The following abbreviations apply: CE – Cartridged Emulsion as the main explosive type in sensitive areas; BE – Bulk Emulsion as the main explosive type in areas away from sensitive receivers. In the CE tunnel type of blasting, bulk emulsion will not be used and marked "N/A" in the table.
2. The pull length for CE is 2m while the pull length for BE is 5.5m.
3. The actual figures are subject to blast design based on blasting assessment.

12.3.3 Types of Explosives

12.3.3.1 Two types of explosives will be used for blasting:

- Initiative explosives: cartridged emulsion explosives / booster, detonating cord and detonators; and
- Blasting explosives: cartridged emulsion explosives (in closed proximity to sensitive receivers) and site sensitised bulk emulsion explosives.

12.3.3.2 The cartridged emulsion and bulk emulsion contain an oxidising agent mainly composed of ammonium nitrate (AN), water, and a hydrocarbon (e.g. fuel oil). The cartridged emulsion may also contain 2-3% aluminium powder (depending on the manufacturer) to increase the explosion temperature and the explosion power.

12.3.3.3 The delivery of explosives and accessories (cartridged emulsion explosives and detonators, etc.) for the Project will be performed by Mines Division from the Government Explosives Depot, using Mines Division's explosives delivery vehicles, to the explosives magazine. The Contractor will be responsible for the transport of explosives from the explosives magazine to the blasting sites. The transport of explosives from the Government Explosives Depot is under Mines Division's purview and falls outside the scope of this EIA, this hazard to life assessment will only focus on the risks due to storage of explosives in the magazine, transport of explosives from the magazine to the blasting sites and also the use of explosives.

12.3.3.4 Bulk emulsion precursor (an oxidizing agent) will be transported to the blasting sites by the appointed third-party supplier. The bulk emulsion precursor will only become classified as an explosive after being sensitised at the blast location or work face,

by the addition of a gassing agent as it is pumped into the blast holes at the excavation face.

12.3.3.5 Detonators are small devices used to safely initiate the blasting explosives in a controlled manner. The detonators commonly used in Hong Kong are of the non-electric type and are initiated by shock tube. Detonators are manufactured with in-built delays of various durations to facilitate effective blasting and allow shots to be initiated at one time but to fire sequentially. Detonating cords are thin and flexible tubes with explosive core. They detonate along its length continually and are suitable for initiating explosives that are detonator sensitive such as the cartridged emulsion and booster. The core of the cord is a compressed powdered explosive which is usually the PETN, and it is initiated by the use of detonator.

12.3.3.6 Cartridged emulsion / booster, detonating cords and detonators will be used to initiate the blast at the work faces depending on the blasting requirements. The primer will comprise one cartridge of emulsion explosives, into which the detonator will be inserted. Small loops of detonating cord will be used to connect all detonators within one delay sector in the blast pattern. Non-electric type detonators initiated by shock tube are commonly used in Hong Kong.

12.3.4 Properties of Explosives

12.3.4.1 Properties of the two types of explosives to be used in this Project are shown in **Table 12.3**.

Table 12.3 Types of Explosives

Type	Function	Use	Example
Initiating explosives	To initiate the main blasting explosives	Initiation of secondary explosives	Cartridged emulsion explosives / booster, detonators, detonating cord
Blasting explosives	Used as the main blasting explosives	General blasting, shattering rock / structures	Bulk emulsion, cartridged emulsion in closed proximity to sensitive receivers

Cartridged Emulsion

12.3.4.2 The cartridged emulsion is packaged in plastic films with the tips clipped at each end to form a cylindrical sausage, or wrapped in waxed paper. It can be used for both priming and full column applications, such as mining, quarrying and general blasting work.

12.3.4.3 Cartridged emulsion is classified as a UN Class 1.1D explosive and S1DG. It has a TNT equivalence of 0.96, i.e. 0.96kg of TNT in 1kg of emulsion.

12.3.4.4 Cartridged emulsion consists of a mixture of oxidisers and fuel. It contains high quantity of water which is typically around 14%. The oxidisers are typically ammonium nitrate, calcium nitrate or sodium nitrate. The fuels are waxes or oils such as diesel fuel. The whole mixture is complete with small amounts of

emulsifiers to keep the water and oil mixture homogeneous. It is detonator sensitive and does not require the use of booster to cause it to detonate.

Bulk Emulsion Precursor

12.3.4.5 Bulk emulsion has a similar composition to cartridged emulsion except that it does not contain aluminium. The bulk emulsion precursor has a density of 1.38-1.40 g/cc. Prior to sensitisation, it is not considered as an explosive, and is classified as UN 5.1 oxidising substance which yields oxygen readily to stimulate the combustion of other materials.

12.3.4.6 Before sensitisation, bulk emulsion precursor is stable under normal conditions and there is no major fire hazard. The oxidising properties of bulk emulsion precursor are considered the major hazard of it, which can cause irritation to eyes and skin. Explosion of bulk emulsion precursor is only possible under prolonged fire, supersonic shock or very high energy projectile impact.

12.3.4.7 Due to its stability under normal conditions, the storage and transport of bulk emulsion precursor will not be further considered in this QRA.

Blasting Explosives

12.3.4.8 Bulk emulsion will be used as the main blasting explosives to excavate rock by rock blasting. It will be manufactured on-site and requires the use of initiating explosives.

12.3.4.9 The bulk emulsion precursor will be sensitised at the blasting site by adding a gassing solution which contains sodium nitrate.

12.3.4.10 The gassing solution will be injected into the bulk emulsion precursor to reduce the density to 0.8-1.1 g/cc at the discharge end of the loading hose. This produces nitrogen gas bubbles that aid the propagation of the detonation wave, and the emulsion is said sensitised. The sensitised emulsion can then be detonated with the assistance of a cartridged emulsion / small booster and a detonator.

12.3.4.11 The bulk emulsion, once being gassed, is classified as UN 1.1D explosive or S1DG. The bulk emulsion blasting explosives once it is mixed should be pumped into and completely fill the blast hole.

Detonating Devices

12.3.4.12 Detonators are small devices used to safely initiate the blasting explosives in a controlled manner. The detonators commonly used in Hong Kong are of the non-electric type and are initiated by shock tube.

12.3.4.13 Detonators are classified as either UN 1.1B, 1.4B or 1.4S, and S1DG. Although detonators contain the most sensitive types of explosives in common use, they are packaged in a manner that no serious effects outside the package if accidentally initiated, this minimises the risk associated with handling and use of the detonators.

12.3.4.14 Detonators are manufactured with in-built delays of various durations to facilitate effective blasting and allow shots to be initiated at one time but to fire sequentially. The delay time of a detonator is controlled by the burning time of a pyrotechnic ignition mixture pressed into a 6.5mm diameter steel tube. This delay element causes the primary explosive to detonate. This in turn causes the secondary

explosive PETN to detonate. The delay time of a detonator is based on the length of steel tube and the compaction of the pyrotechnic mixture. In designing the blasting of a blasting face, the general principle is to select the required detonators to ensure that each individual detonating blast hole is separated by a minimum of 8ms.

12.3.4.15 Detonating cords are thin and flexible tubes with explosive core. They detonate along its length continually and are suitable for initiating explosives that are detonator sensitive such as the cartridged emulsion. The core of the cord is a compressed powdered explosive which is usually the PETN, and it is initiated by the use of detonator.

12.3.5 Charges and Delays

12.3.5.1 In order to facilitate a smooth blasting, perimeter holes will be lightly charged, using either detonating cords or string loading of emulsion. Detonating cords must be delivered to site as Schedule 1 Dangerous Goods (S1DG) (explosives). The typical quantity used per perimeter blasthole is two strips of 40g per meter detonating cord.

12.3.5.2 Each blasthole will also require a detonator, and it is also delivered to site as S1DG (explosives). Each detonator typically contains up to 1g of explosive as the base charge. A small quantity of detonators or light (5g per meter) detonating cord is also required to complete the hook-up of all the individual blasthole detonators.

12.3.5.3 The cartridged emulsion / boosters, detonating cord and detonators would need to be delivered to site as S1DG (explosives). These are commonly all called packaged explosives, to distinguish them from bulk emulsion, which is not a S1DG (explosives) when stored or delivered to site.

12.3.5.4 It is assumed that where the MIC is restricted to less than 1 TNT eqv. kg, cartridges initiated by electronic detonators would be selected to provide additional confidence in the MIC per delay and remain within the quoted PPV limits. Where MIC is greater than 1 TNT eqv. kg, it is expected that the bulk emulsion will be charged with a single 200g emulsion cartridge or a mini cast booster per blasthole to initiate the bulk emulsion.

12.3.5.5 Bulk emulsion explosives, which will only being sensitised on site, will be adopted as far as practicable to reduce the number of explosive delivery trips. Cartridged emulsion, which relay on the delivery from the Government Explosives Depot, will be adopted for better chargeweight control when tunnel blasting close to sensitive receiver which the allowable chargeweight is low.

12.3.5.6 A majority of drill-and-blast tunnels for the Project are expected to be excavated using bulk emulsion to be initiated by non-electric detonators. It is assumed in this assessment that the MIC for any single detonation will be 10 TNT eqv. kg. for all tunnel blasting.

12.3.6 Magazine Requirement

12.3.6.1 To allow flexibility and secure the programme for excavation operation during construction phase, a fixed time once per day explosives delivery arrangement by Mines Division may not suit the need of the project. Therefore, an explosive magazine is required. Apart from construction phase, explosives are not expected to be used, stored or transported, during operation phase of the Project.

12.3.6.2 The proposed temporary explosive magazine site at Tai Shu Ha (Yuen Long) in fact is the former magazine site of XRL Project and CEDD Liantang projects. Besides, the site will be occupied by the MTRC Northern Link Project. Close liaison with the MTRC Northern Link project team will be kept for the progress of blasting activity of the Northern Link project and hand over of the Tai Shu Ha magazine site to this Project.

12.3.6.3 The explosives storage of the proposed magazine site is limited to 800 TNT eqv. kg (with two 400 TNT eqv. kg storages) and the explosives stored would include initiating explosives and blasting explosives.

The Need of a Magazine

12.3.6.4 Daily blasting will be anticipated for tunnel excavation throughout the project. The supply of explosives for blasting relies on a daily delivery from Mines Division. This delivery is typically made available between 11am and 2pm each day, and the explosives are expected to be used immediately, limiting to a single blast each day during the afternoon. In order to expedite the tunnel excavation with minimised idle time in the blast cycle and mitigate the potential risk of missing out the explosives delivery by Mines Division due to unforeseen site conditions e.g. extra time for temporary support, machine malfunction, an explosive magazine with storage capacity of 800 TNT eqv. kg at the magazine site is required to allow flexible blasting time.

12.3.6.5 It shall be noted there may be other situations which might require additional explosives quantities, such as areas of low Maximum Instantaneous Charge (MIC), less than 1 TNT eqv. kg allowable MIC, or additional blasting for connecting passages, niches etc. which are in addition to the regular blast faces for tunnels.

Selection of Magazine Site

12.3.6.6 Following the searches for potential magazine sites for previous Drill-&-Blast tunnel projects, a further review has been carried out for the Project. Six potential magazine sites have been identified, including three previously used magazine sites (Tai Shu Ha, So Kwun Wat and Sha Tin Cavern magazine site) for the Hong Kong Section of Guangzhou – Shenzhen – Hong Kong Express Rail Link (XRL), CEDD Liangtang projects and Drainage Services Department Shatin Cavern project, three other potential magazine sites at Ngau Kwu Leng Trail, Firing Range and Lam Tei Quarry. **Appendix 12.9** summarizes the potential magazine site selection.

12.3.6.7 Tai Shu Ha (Yuen Long) has been chosen as the proposed temporary magazine site due to its better suitability in terms of land availability, relatively less public disturbance, remote location, and compliance with established design standard. An additional benefit of selecting this location is its close proximity to the proposed blasting sites, which reduces transportation distance and time for delivering explosives to the construction site.

12.3.6.8 Temporary explosive magazine site at Tai Shu Ha (Yuen Long) is anticipated to be utilized by NOL project before it can be used by NMH project. Liaison with NOL project for the handover of magazine site is required.

12.3.6.9 The site was previously used as the magazine site for the XRL Project and CEDD Liantang projects but all magazine structures and ancillary facilities have already been demolished.

12.3.6.10 The UK Explosives Regulations 2014 [35] specified the external separation distances between an explosives store and neighbouring landuse, based on the quantity of explosives for each individual store where store means a building, enclosed area or metal structure in which explosives are, or are to be, stored. For the proposed magazine site used in this Project, the explosives would be stored in two brick-built mounded stores with a capacity of 400kg each, a separation distance of 183m should be kept from nearest building and major road. The proposed magazine site is situated in a relatively remote area. The nearest village house is located about 300m away and the nearest building of the Hong Kong Model Engineering Club (HKMEC) is about 250m away, meeting the required separation distance criteria. Although the HKMEC boundary lies around 160m away from the proposed magazine site, its operator periodically flies model aircrafts at the runway at about 200m away from the site which is far from the proposed magazine site.

12.3.7 Statutory / Licensing Requirement and Best Practice

Storage of Explosives

12.3.7.1 Explosives are classified as Class 1 Dangerous Goods, and the Commissioner of Mines shall have the control and management of every depot subjected to Section 13C of Dangerous Goods Ordinance (Cap. 295). The storage of explosives for blasting must be within a licensed Mode A Store. All the safety and operational criteria must be met to get the approval of Mines, CEDD.

12.3.7.2 The general requirements stated in the “Guidance Note No. GN8 How to Apply for a Mode A Store License for Storage of S1DG (Blasting Explosives)” must be met to obtain a licensed Mode A Store for the magazine site, including but not limited to:

- The storage capacity should be kept to the minimum required for the project to minimise risk to the public;
- The proposed magazine should be located on plan at least 45m and 75m from any high tension power cables carrying 440V and 1KV respectively;
- The security aspects of the Mode A store location and the security company should be approved by the Commissioner of Police; and
- Other materials likely to cause fire or explosion should not be transported in the explosives delivery vehicles, and only the persons assigned to assist in handling explosives should be permitted on an explosives delivery vehicle. Driver and all workers engaged in the loading, unloading and conveying of explosives should be trained in firefighting and precautions for the prevention of accident by fire or explosion.

12.3.7.3 The general requirements for the construction of an explosives Mode A store magazine are listed as follows:

- The store should be a single storeyed detached structure with lightning protection and outer steel store doors;

- All hinges and locks should be made of non-ferrous metal;

- No ferrous metal should be left exposed in the interior of the Mode A store;

- The interior and exterior walls of the Mode A store should be painted white;

- The outer side of the steel door of the Mode A store should be painted red. The words "DANGEROUS – EXPLOSIVES" and "危險品- 爆炸" should be written in white on the outside of the door. The letters and characters should be at least 100mm high. No ferrous metal shall be exposed on the inner face of the door forming part of an interior of the Mode A store;

- A security fence surrounding the Mode A store should be installed and set back at least 6m from the store. The fence should be 2.5m high, stoutly constructed of chain link fencing with a mesh size not exceeding 50mm. the fence should be firmly fixed to metal or concrete posts and topped with a 0.7m high outward overhang of razor wire. The base of the fence located between the posts should be secured with pegs to prevent intrusion;

- The area between the security fence and the Mode A store should be cleared of all vegetation. Vegetation clearance should also apply to a minimum distance of 1m on the exterior of the fence. A uniform cross-fall of at least 1 in 100 away from the store to a drainage system should be constructed;

- The road leading to the Mode A store should have a concrete surface and it should be constructed and maintained so that 11 tonne trucks can use it under all weather conditions. A suitable turning circle or other alternative means for these trucks to turn should be provided so that the trucks can be driven up to the gate of the security fence;

- The gate in the security fence should be fitted with a lock of close shackle design with a key-retention feature. A warning signboard with prohibited articles and substances painted in red and black, shown in symbols and in Chinese and English characters should be posted at the gate. Each symbol should be at least 100mm in diameter.

- A guardhouse should be provided. For surface Mode A store, security guards should be on duty outside the inner security fence adjacent to the fate when there is no receipt or issue of explosives inside the Mode A store. A separate outer security fence should be installed to protect this guardhouse;

- Inside the guardhouse, an arms locker constructed as an integral part of the house and fitted with a lock is required;

- A telephone should be provided for use by the guard in the guardhouse. A watchdog should normally be provided for the store; and

- Fire-fighting installations consisting of at least four 6 litre foam and one 4.5kg dry powder fire extinguishers to be positioned on two racks and four buckets of

sand should be provided at the nearest convenient locations to the Mode A door.

Transport of Explosives

12.3.7.4 The explosives delivery vehicles must comply with the regulations as stated in the "Guidance Note No. GN2 Approval of an Explosives Delivery Vehicle" published by the Mines Division.

12.3.7.5 The safety requirements for approval of an explosives delivery vehicle and requirements for signage on the truck are listed as follows:

- Condition of Vehicle
  - The vehicle should be powered by a diesel engine;
  - The vehicle's design, construction and strength must comply with the Road Traffic (Construction and Maintenance of Vehicles) Regulations, Chapter 374; and
  - The vehicle should be kept clean, in sound mechanical condition and roadworthy.
- Condition of Cargo Compartment
  - The cargo compartment including the roof should be constructed with sheet metal at least 3mm thick and lined internally with at least 13mm thick plywood, and there should be no exposed ferrous metal in the interior of the goods compartment;
  - The interior of the cargo compartment including doors should be kept in good condition and free from defects or projections which might cause accidental damage to the packages;
  - Electric wiring or electrical devices should not be installed inside the cargo compartment;
  - The door of the cargo compartment should be capable of being locked;
  - Proper stowage facilities should be provided to secure the load in a stable manner during transportation; and
  - If the vehicle is designed to carry both detonators and other types of blasting explosives at the same time, additional requirements, given in AEISG (2014) [32], are required.
- Safety Provisions
  - The driver's cabin should be separated by a distance of at less than 150mm from the cargo compartment of the vehicle;
  - The exhaust system must be located as far from the cargo compartment as possible, preferably at the front of the vehicle;
  - An emergency fuel cut-off device should be located at an easily accessible position with a label, in Chinese and English, prominently and legibly stating: "EMERGENCY ENGINE STOP 緊急死火掣"
  - For a typical vehicle with gross vehicle weight of 9 tonnes or above, four fire extinguishers, comprising two 2.5kg dry powder and two 9-litre foam

fire extinguishers of an approved type, with certificates, shall be provided. They shall be mounted in front and on both sides of the rear body in easily accessible positions with securely mounted brackets and quick release clamps;

- All electrical installations shall be designed, constructed and protected so that they cannot cause any ignition or short-circuit under normal conditions of use of the vehicle or its electrical installations, and to ensure that the risk of this occurring will be minimized in the event of a traffic accident. All electrical wiring and fittings shall be shrouded in fire resisting conduits;
- The fuel tank shall be located below the cargo compartment of the vehicle. It shall be protected from accidental damage and designed to prevent accumulation of spilt fuel on any part of the vehicle;
- Fire resistant material shall be fitted between the wheel arches and the cargo compartment;
- Detonators and other types of blasting explosives shall not be loaded or transported within the same cargo compartment of the vehicle, unless the cargo compartment fulfils the additional requirements as specified in Annex B of "Guidance Note on Requirement for Approval of an Explosives Delivery Vehicles";
- A hand-held lightning detector shall be provided in the vehicle for detection of lightning before and during loading and unloading of explosives. Should lightning be detected within a distance of 16 km from the loading/unloading point by the hand-held detector, loading or unloading of explosives shall cease until the lightning signal has cleared;
- A fire suppression system must be fitted to the engine bay of vehicles, complying with the AS5062-2016 [33]; and
- For a typical vehicle with a gross vehicle weight of 9 tonnes or above, two strobe beacons approved by the Transport Department must be installed on top of the cargo compartment. For a vehicle with gross vehicle weight of less than 9 tonnes, at least one strobe beacon is required.

- Signage on vehicles
  - Whenever the vehicle is carrying explosives, it shall display: (i) on both sides and on the rear door of the cargo compartment, placards (of minimum dimensions 250mm x 250mm) showing the label of the highest Hazard Code of explosives, and (ii) a rectangular red flag, in a prominent position, of dimensions not less than 230mm x 300mm;
  - Placards showing "EMPTY 空車" or blank placards shall be displayed when the vehicle is empty; and
  - The vehicle should be printed in white with warning words in the Chinese and English of at least 150mm height. The word "DANGER-EXPLOISVES" and "危險-爆炸品" should be printed in red colour and displayed on both sides and rear face of the goods compartment.

Use of Explosives

12.3.7.6 Bulk emulsions and bulk Ammonium Nitrate / Fuel Oil (ANFO) are commonly manufactured at the blast sites and being used immediately for rock blasting. Under Section 16 of the Dangerous Goods (Control) Regulation, Cap. 259G, a manufacture (blasting) license is required to manufacture nitrate mixture, i.e. Group 2 Schedule 1 Dangerous goods (S1DG), within a blasting site. The Commissioner of Mines is the Authority for issuing the license.

12.3.7.7 For the manufacturing of bulk emulsion at blast sites, ammonium nitrate (AN), which is classified as DG Class 5.1 – Oxidizing Substances under Schedule 2 of the Dangerous Goods (Application and Exemption) Regulations Cap. 295E, is used. A licence for the storage of DG Class 5 is required according to "A Guide to Application for Dangerous Goods Licence" from the Fire Services Department.

12.3.7.8 For the use of explosives, a blasting permission is required from the Mines Division so that the use of explosives at a work site for the carrying out of blasting is allowed; and a Mine Blasting Certificate is required so that the shotfirer is permitted to conduct blasting operations.

12.3.8 Construction Cycle and Programme

12.3.8.1 After commissioning of the magazine the proposed delivery-storage-blasting cycle will consist of the following elements:

- Weekdays and Saturdays morning deliveries of explosives and initiating systems to magazine by Mines Division as needed;
- Storage in the magazine stores;
- Transfer from the magazine stores to the explosives unloading points utilising public roads, the transport routes are shown in **Appendix 12.1**;
- Transfer to the working faces of the excavation; and
- Load and fire the faces to be blasted. Blasts in a particular area will be initiated from a common firing point once all personnel are cleared and entry routes to each blast site are secured.

12.3.9 Explosives Transport Requirements

12.3.9.1 The approach adopted to derive the total number of trips and the total initiating explosives to be transported per trip is as follow:

- As far as practicable, the explosives (i.e. cartridged emulsion and detonating cord) required for all blast faces of a given work area operated by the same contractor will be transported on the same explosive delivery truck when the blasting programmes for the blast faces of the work area overlap. Detonators are transported on a separate explosives delivery vehicle at the same time;
- Since potential progress issues may be arising during the construction stage which may cause the delay or change of programme, it may not be possible to adhere strictly to the envisaged construction programme which may result in difference of the actual blasting time and separate deliveries of explosives;

- The quantity of Class 1 explosives on the roads has been minimised by using bulk emulsion and/or ANFO, which will be manufactured on-site. The on-site manufacture of ANFO and bulk emulsion will require the transportation of Class 5.1 oxidising substances which falls outside the scope of the QRA;

- It is assumed that the project will mostly require a separate explosive delivery from the relevant explosives unloading points to each tunnel portal; and

- The actual construction programme will depend on the detailed design and contractors. It may also depend on the actual achievable progress rates which may vary due to specific site conditions.

12.3.9.2 In the Base Case, it was also considered that blasting could be carried out at predetermined time during the day as given in the designed construction programme. A distribution of delivery time has thus been considered based on the envisaged construction programme. The details of transportation for the Base Case Scenario are summarised in **Table 12.4**.

12.3.9.3 This assessment also covers the Worst Case scenario that there is a possibility for the construction programme to be different from the envisaged one due to construction uncertainties or contractor's method of working, which may increase the number of delivery trips and return trips, or the number of blasts per day etc. The QRA results for the Base Case scenario will be increased by 100% to represent the Worst Case scenario. The assumption of 100% increase for Worst Case scenario is very conservative and would only happen in unusual situation. The details of transportation for the Worst Case scenario are summarised in **Table 12.4**.

Table 12.4 Details of Explosives Transport

From	To	Base Case Scenario		Worst Case Scenario	
		Volume (TNT eqv. kg per trip)	No. of Transport (trips per year)	Volume (TNT eqv. kg per trip)	No. of Transport (trips per year)
Magazine	Explosives Unloading Point I	200	173	200	346
		130	249	130	498
Magazine	Explosives Unloading Point II *	200	202	200	404
		140	333	140	666
Magazine	Explosives Unloading Point III #	200	139	200	278
		60	100	60	200
Explosive Unloading Point I	Tunnel A South Portal	200	173	200	346
		130	249	130	498
Explosive Unloading Point II	Tunnel A North Portal	200	43	200	86
		140	275	140	550
	Tunnel B South Portal *	200	159	200	318
		140	58	140	116
	Tunnel E East Portal *	200	50	200	100
	Tunnel E West Portal *	200	50	200	100
Explosive Unloading Point III	Tunnel B North Portal #	200	139	200	278
		60	100	60	200
	Tunnel C South Portal #	200	40	200	80
		60	38	60	76
	Tunnel C North Portal #	200	40	200	80
		60	38	60	76

Remarks:
* Blasting of Tunnel E (from both East Portal and West Portal) will not be carried out concurrently with Tunnel B (from South Portal).
Blasting of Tunnel C (from both South Portal and North Portal) will not be carried out concurrently with Tunnel B (from North Portal).

12.3.10 Concurrent Project during Construction Phase

12.3.10.1 The magazine site will be handed over from Northern Link project and thus no construction of the magazine is required under this Project, while the magazine site may be used for the construction of other sections of the NM Highway and thus no decommissioning of the magazine is required under the San Tin Section. With a view to assessing the risk impact on all the potential sensitive receivers, and to assess any potential cumulative risk impact of the Project, the concurrent projects identified in Section 2 is summarised in **Table 12.5** based on the best available information at the time of preparation of this report. Locations of these identified concurrent projects please refer to **Figure 2.3**.

Table 12.5 Summary of Concurrent Projects

Project	Tentative Construction Programme	Possible Cumulative Impacts
Kwu Tung North New Development Area	2013-2031	The separation distance between the proposed "Kwu Tung North New Development Area" and the blasting sites under the currently proposed project is over 400m, no cumulative risk impacts during construction are anticipated.
Ngau Tam Mei Water Treatment Works Extension	2025-2030	The extension works will be completed during the assessment year of blasting under the currently proposed project, staffs of the water treatment works during its operation phase are considered as the risk receivers.
Ngau Tam Mei New Development Area	2027-2036	The construction works of the proposed "Ngau Tam Mei New Development Area" will be concurrent with the blasting activities under the currently proposed project, construction workers are considered as the risk receivers.
Northern Metropolis Highway (NMH) – Kwu Tung Section	Under review, target completion in 3-4 years after NMH-San Tin Section commissioning	No concurrent construction of both projects, no cumulative risk impact is anticipated.
Northern Metropolis Highway (NMH) – Tin Shui Wai Section	Under review, target completion in 3-4 years after	No concurrent construction of both projects, no cumulative risk impact is anticipated.

Project	Tentative Construction Programme	Possible Cumulative Impacts
	NMH-San Tin Section commissioning	
Northern Link (NOL)	2025-2034	There will be no concurrent use of the magazine and thus no cumulative risk associated with the storage and transport of explosives. The blasting location of 2 projects are about 2km away from each other and therefore no cumulative risk associated with the use of explosives is anticipated.
Planned Residential Development under Planning Application no. Y/YL-NSW/7	2026-2030	Construction works for the "Planned Residential Development under Planning Application no. Y/YL-NSW/7" will be completed during the assessment year of blasting of the currently proposed project, residential population will be considered as the risk receivers.
Planned Residential Development under Planning Application no. Y/YL-NSW/8	Completion in 2031	Construction works for the "Planned Residential Development under Planning Application no. Y/YL-NSW/8" will be completed during the assessment year of blasting of the currently proposed project, residential population will be considered as the risk receivers.
Planned Residential Development under Planning Application no. Y/YL-NSW/9	Completion in 2031	Construction works for the "Planned Residential Development under Planning Application no. Y/YL-NSW/9" will be completed during the assessment year of blasting of the currently proposed project, residential population will be considered as the risk receivers.
Planned Residential Development under Land Sharing Pilot Scheme Application No. LSPS/002	Completion in 2031	Construction works for the "Planned Residential Development under Planning Application no. LSPS/002" will be completed during the assessment year of blasting of the currently proposed project, residential population will be considered as the risk receivers.

Project	Tentative Construction Programme	Possible Cumulative Impacts
Planned Residential Development under Planning Application no. A/YL-NSW/274	No publicly available information	Project details and construction programme are unavailable, the proposed development will not be considered.
Planned Residential Development under Planning Application no. A/YL-NSW/314	No publicly available information	The site is located outside the assessment area of the transport of explosives under the currently proposed project, and it will not be considered.
San Tin Technopole	2024-2039	Construction works of the proposed "San Tin Technopole" will be overlapped with the blasting activities of the currently proposed project, construction workers are considered as the risk receivers.
Site Formation and Infrastructure Works for Public Housing Development at Sha Po, Shap Pat Heung and Tai Kei Leng Yuen Long	Under review by the Government	Project details and construction programme are unavailable, the proposed development will not be considered.
Proposed Nam Sang Wai Wetland Conservation Park	Under review by the Government	Project details and construction programme are unavailable, the proposed development will not be considered.
Planned Residential Development Under the Planning Application no. A/YL-NSW/293	No publicly available information	Project details and construction programme are unavailable, the proposed development will not be considered.
Planned Residential Development Under the Planning Application no. A/YL-KTN/964	No publicly available information	Project details and construction programme are unavailable, the proposed development will not be considered.
Planned Residential Development Under the Planning Application no. A/YL-KTN/604 and A/YL-KTN/663	Completion of Phase A in 2023; No publicly available information of	The site is located outside the assessment area of the transport of explosives under the currently proposed project, and it will not be considered.

Project	Tentative Construction Programme	Possible Cumulative Impacts
	Phase B development	
Planned Social Welfare Facilities under the Planning Application no. A/YL-NSW-348 and A/YL-NSW-349	Completion in 2030	The site is located outside the assessment area of the transport of explosives under the currently proposed project, and it will not be considered.
Planned School under the Planning Application no. Y/NE-KTS/18	Completion in 2036	The site is located outside the assessment area of the transport of explosives under the currently proposed project, and it will not be considered.

12.4 Study Approach

12.4.1.1 The elements of the QRA are shown schematically in **Diagram 12-3**. It consists of the following 6 main tasks:

i. Data / Information Collection and Update: Collect and review relevant data associated with the storage, transport and use of explosives at the works areas, as well as population and vulnerable receptors, such as slopes, retaining walls etc., in the vicinity of blasting activities.

ii. Hazard Identification: Identify hazardous scenarios associated with the storage, transport and use of explosives.

iii. Frequency Estimation: Estimate the frequencies of each hazardous event leading to fatalities with full justification by reviewing historical accident data and previous similar projects.

iv. Consequence Analysis: Analyse the consequences of the identified hazardous scenarios.

v. Risk Integration and Evaluation: Evaluate the risks associated with the identified hazardous scenarios. The evaluated risks will be compared with the HKRG to determine their acceptability. Where necessary, risk mitigation measures will be identified and assessed to comply with the ALARP principle used in the HKRG.

vi. Identification of Mitigation Measures: Review the recommended risk mitigation measures from previous studies, practicable and cost-effective risk mitigation measures will be identified and assessed as necessary. Risk outcomes of the mitigated case will then be reassessed to determine the level of risk reduction.

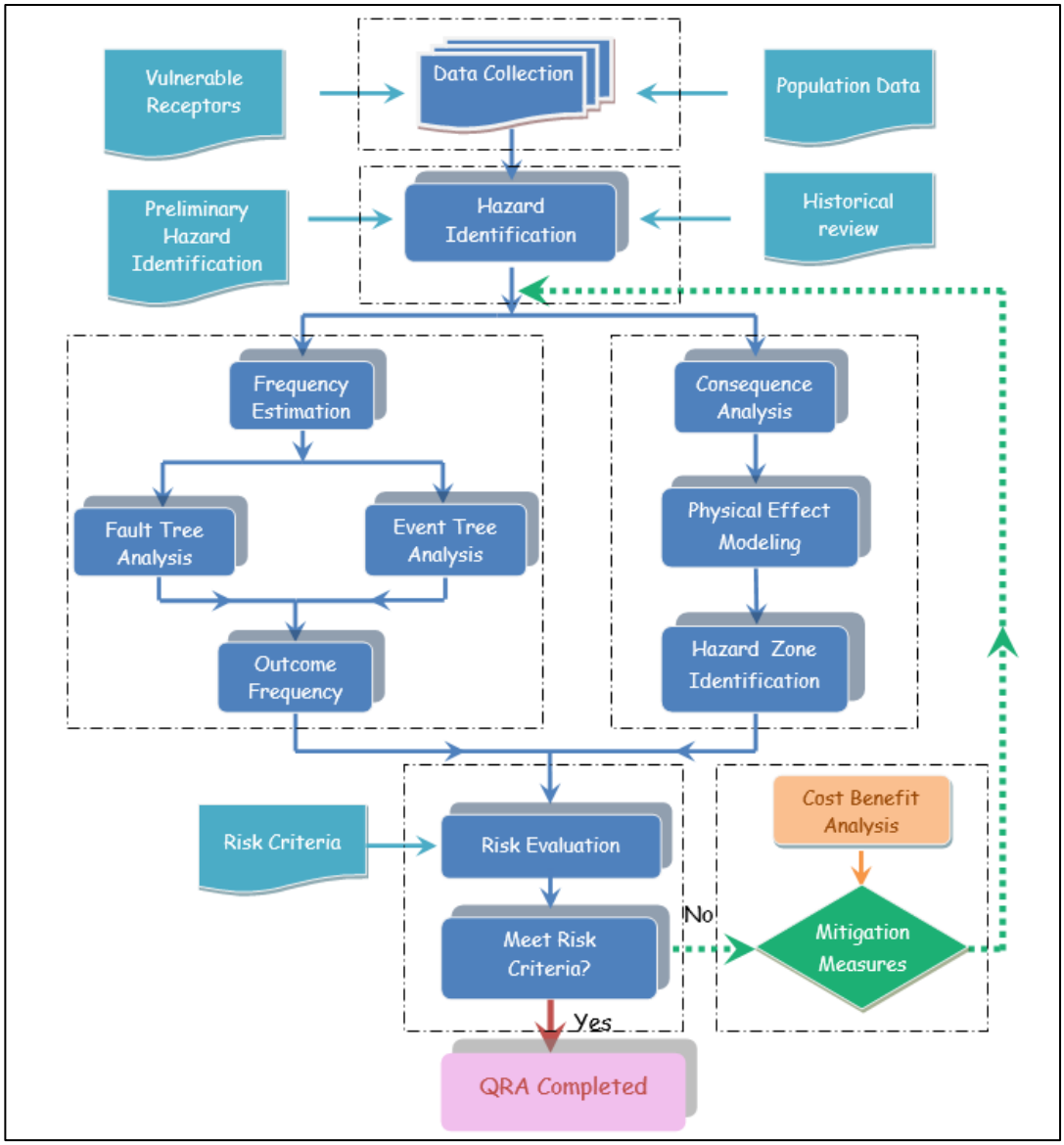


Diagram 12-3 Schematic Diagram of QRA Process

12.4.1.2 This hazard to life assessment covered the following scenario:

i. Year 2033 – The risk imposed by the storage, transport and use of explosives to the existing, committed and planned population in 2033.

12.5 Estimation of Population

12.5.1 Assessment Area

12.5.1.1 Taking into account the affected distances of hazards associated with the transport and use of explosives (i.e. 1% fatality probability of blast effects as shown in **Table 12.20**, and the 90 mm/s screening criteria for slopes and boulders), all off-site population located within an assessment area comprising a 100m influence zone from the tunnel portals, along the tunnels, and also from the magazine and along the transport routes is considered.

12.5.2 On-site Population

12.5.2.1 The risk guidelines specified in the EIAO-TM apply to the risk of fatality of the public outside the boundary of the hazardous installation (i.e. the Project Boundary). The construction workers deployed to the project work sites during the construction phase are regarded as voluntary risk takers and referred to on-site population in the QRA.

12.5.3 Off-site Population

12.5.3.1 Three types of population were considered within the influence zone of the tunnel blasting and also along the transport route of explosives:

- Land and building population;
- Road population; and
- Pedestrian population.

12.5.3.2 The locations of populations considered in this QRA are shown in **Appendix 12.2**.

Land and Building Population

12.5.3.3 Lands and buildings located within the influence zone of tunnel blasting as well as those along the transport routes of explosives were included in this QRA. The population data was estimated and projected to the assessment year based on the following information:

- 2021-based Territorial Population and Employment Data Matrix (TPEDM) by Planning Department;
- 2021 Population Census;
- Annual reports of relevant schools; and
- Code of Practice for Fire Safety in Buildings 2011 (version September 2024);
- Hong Kong Planning Standards and Guidelines (HKPSG); and
- EIA reports of relevant developments.

12.5.3.4 Residential population of the existing buildings in the assessment year was estimated based on the average household size obtained from the 2021 Population Census. According to the census data, the average household size in Yuen Long district was 2.8. A decreasing trend of the household size from 2.8 in 2021 to 2.7 in 2036 was shown in the latest TPEDM data. To be conservative, the residential population in the assessment area in 2033 was estimated based on the average household size of 2.8.

12.5.3.5 The blasting activity of this project will be concurrent with the construction phase of both the San Tin Technopole and Ngau Tam Mei Development. According to the EIA studies of both developments [29][30] , the maximum number of construction workers in the 610ha San Tin Technopole construction site will be no more than 3,000; while the maximum number of construction workers in the 130ha Ngau Tam

Mei Development construction site will be no more than 870. This equivalent to 0.0005 construction worker per m^2 in the San Tin Technopole construction site and 0.0007 construction worker per m^2 in the Ngau Tam Mei Development construction site.

12.5.3.6 The number of population in each population polygon considered in this QRA and the information sources are provided in **Appendix 12.2**.

Road Population

12.5.3.7 The traffic data was based on the latest Annual Traffic Census (ATC) [31] published by Transport Department (TD) and the information provided by the Traffic Consultant. The road population density was calculated using the following equation:

$$\text{Population density (persons/m}^2\text{)} = \frac{\text{Traffic flow (no. of vehicle/hour)} \times \text{Occupancy (no. of person/vehicle)}}{\text{Width of road (m)} \times \text{Vehicle speed (km/hr)} \times 1000}$$

12.5.3.8 The impact distances of blast effect due to detonation of explosives are larger for indoor population than those for outdoor population (details please refer to **Section 12.8**). To be conservative, the population associated with road vehicles was modelled as 100% indoor.

12.5.3.9 The number of daily traffic and vehicle mix is taken from the traffic forecast provided by the Traffic Consultant while the occupancies for each vehicle type were taken from the ATC for year 2024. Details please refer to **Appendix 12.2**.

Pedestrian Population

12.5.3.10 All footpaths located within the assessment area (**Appendix 12.2** refers) were considered.

12.5.3.11 Pedestrian population of the pavements along the Kwu Tung Road and Kwu Tung South Road, and also those along the identified footpaths within the assessment area are presented in the form of population density. Site survey supplemented with desktop search of approved EIA reports found that the pedestrian population density of the pavements / footpaths within the assessment area of this QRA were similar or less than those considered in the Sha Tin Cavern EIA[1]. The pedestrian population density of all pavements / footpaths considered in this QRA was conservatively assumed to be the same as those considered in the Sha Tin Cavern EIA, i.e. 0.000428 person per m^2 at any time.

12.5.4 Time Periods and Occupancy

12.5.4.1 Four time-modes were adopted to take into account the temporal changes in population within the assessment area. Detailed descriptions and time fraction of each time-mode are presented in **Table 12.6**. The temporal population distribution makes reference to the QRA of approved EIA with similar issues such as XRL EIA,

[1] AECOM, 2016. EIA Study on Sha Tin Cavern Sewage Treatment Works: Hazard to Life Assessment. (Register No.: AEIAR-202/2016)

Sha Tin Cavern EIA, Route 11 EIA, etc. The temporal population distribution of various time modes is shown in **Table 12.7**.

Table 12.6 Definition of Time Modes

Time Modes	Description	Period	Weighting per week	Time Fraction
TM1	Weekday Day	Monday to Friday (09:00-18:00)	45 hours	0.268
TM2	Peak Traffic Hours	Monday to Sunday (07:00-09:00 and 18:00-20:00)	28 hours	0.167
TM3	Weekend Day	Saturday and Sunday (09:00-18:00)	18 hours	0.107
TM4	Night	Monday to Sunday (20:00-07:00)	77 hours	0.458

Table 12.7 Temporal Population Distribution

Type	Weekday Day	Peak Traffic Hours	Weekend Day	Night
Residential	20%	50%	80%	100%
Educational	100%	10%	55%	0%
Temple	50%	10%	100%	0%
Recreational / Open Space	70%	10%	100%	10%
Office (Administration) / Commercial	100%	10%	100%	10%
Industry	100%	10%	55%	10%
Open Storage	100%	1%	51%	0%
Car Park	70%	100%	70%	10%
Petrol Station	50%	100%	50%	50%
Road traffic population [1]	Various			
Pedestrian	100%	100%	100%	100%
Construction Site [2]	100%	10%	55%	10%

Note:
[1] The temporal population distribution of road traffic population was based on the traffic data forecasted by engineers for the study.
[2] To allow flexibility of construction arrangement in the Northern Metropolis, no temporal variation of the construction workers in the construction sites of Ngau Tam Mei Development and San Tin Technopole were considered in this QRA, i.e. 100% of maximum population in all time modes was adopted in the assessment.

12.5.5 Features Considered in this Study

12.5.5.1 A number of features including man-made slopes, retaining walls, building structures and utilities were identified within the assessment area. These features could be affected by secondary or tertiary hazards associated with the transport and use of explosives.

Man-made Slopes and Retaining Walls

12.5.5.2 These features included cut slopes, fill slopes, retaining walls and a combination of all the above. These slopes are covered with all types of facing, including shotcrete, chunam, stone facing and vegetation. The slope features considered in this assessment are shown in **Appendix 12.3**.

Natural Terrain Hillside and Boulders

12.5.5.3 The blasting sites are surrounded by natural terrains, all the natural terrains located within the assessment area were considered in this QRA.

Existing Buildings and Structures

12.5.5.4 Buildings and structures located within the assessment area were considered in this QRA.

Utilities

12.5.5.5 There is an underground high-pressure town gas transmission pipeline (HP pipeline) running along the Kwu Tung Road, San Tin Road and Yuen Long Highway, and it is located more than 300m away from the nearest blasting site, i.e. Tunnel C, it is not anticipated to be affected by the blasting works. However, an interfacing section of around 9.5km between the HP pipeline and the explosives transport route is expected. With the consideration than the explosives transportation is a transient event and aboveground, and the HP pipeline is buried underground with a minimum of 1.1m ground cover, the potential leakage of the HP pipeline due to an accidental explosion during the explosives transport is considered to be negligible. Details please refer to **Section 12.7.5**.

12.5.5.6 The Ngau Tam Mei Water Treatment Works (NTM WTW) is located near the South Portal of Tunnel A (**Diagram 12-4**). Several dangerous goods are stored within NTM WTW, potential impact of the dangerous goods storage due to transport of explosives and also the blasting activities were evaluated. The facilities in the NTM WTW involving dangerous goods storage include the chlorine building, ozone building, liquid oxygen tanks and the chemical building, among which the chemical building is the closest to both the transport route of explosives and also the proposed blasting for Tunnel A, with separation distances of around 180m and 190m, respectively. Based on the analysis in **Section 12.8.5**, the dangerous goods storage will not be affected by either primary, secondary or tertiary effects due to the transport of explosives and blasting works at such separation distances.

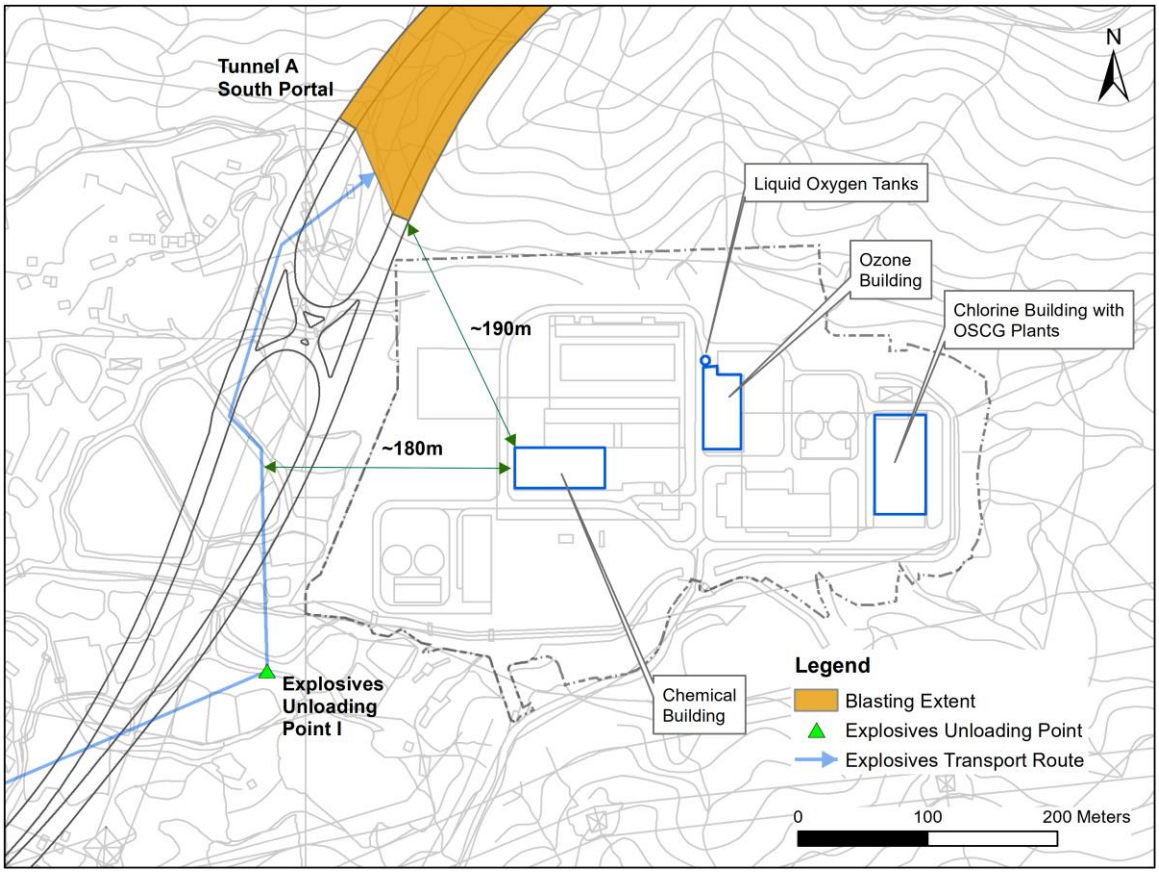


Diagram 12-4 Layout Plan of NTM WTW

12.6 Hazard Identification

12.6.1.1 Hazard identification consists of a review of the following:

- Properties of the explosives;
- Historic accidents;
- Scenarios presented in previous similar studies; and
- Discussion with blasting specialists.

12.6.2 Accidental Initiation due to Hazard Properties of Explosives

12.6.2.1 The types and properties of explosives commonly adopted for blasting in Hong Kong are shown in **Table 12.8** below.

Table 12.8 Types and Properties of Explosives

Explosives Type	TNT Equivalent	Melting Point at 1atm, °C	Auto-ignition Point at 1atm, °C	UN Hazard Division
Cartridged Emulsion	0.96	170	230 - 265	1.1D
Cast Booster	1.3	80	180-190	1.1D
PETN (for detonating cords)	1.4	135 - 145	190	1.1D
PETN (for detonators)	1.4	120	190	1.4B / 1.4S

12.6.2.2 Explosives can be ‘initiated’ by a self-sustaining exothermic reaction. Explosives initiation can result in a vigorous burning without progression to explode, a deflagration or a detonation. It is noted that a deflagration may transit to a detonation and the corresponding mechanism is still under research. Travelling speed of the flame front is the major difference between deflagration and detonation, deflagration produces a subsonic flame front while detonation produces a supersonic one. However, either kind of explosion can lead to severe fatalities and hence should be considered in the QRA.

12.6.2.3 The most common scenario of accidental initiation is basically the cause of fire. Other scenarios of accidental initiation include severe impact and friction. In general, an event with casualty concerns should be at least a deflagration. To induce a deflagration, the explosives should be, at least but not only, exposed to the following stimulus:

- Local stimulus: to generate a ‘hot spot’ like sparks, friction, impact, static electricity, etc.;
- Shock stimulus: shock or high velocity impact such as bullet impact, detonation of other explosives, etc; or

- Thermal stimulus: intense heat or fire. It can be assumed that there can be no significant event until the medium becomes molten. In the case of the emulsion, there can be no significant event until much of the water has been lost.

12.6.2.4 However, not all of these causes are necessarily leading to a deflagration or detonation of the explosives.

12.6.2.5 Accidental initiation of explosives has been categorized as either fire or non-fire induced in this QRA.

Hazard Properties of Emulsion Types Explosives

12.6.2.6 Typical emulsion explosives contain more than 78% Ammonium Nitrate (AN), which is considered as a powerful oxidizing agent. Friction or impact found in normal handling will not trigger initiation of emulsion based explosives. However, heat and confinement or severe shock (e.g. from other explosion) can cause explosion of them. The sensitivity of AN based explosives to deflagration or detonation is proportional to temperatures.

12.6.2.7 There are two broad categories of emulsions:

- Packaged emulsion (sensitized); and
- Bulk emulsion precursor (void-free liquid)

12.6.2.8 Cartridged emulsions are sensitized by either adding gassing solution or plastic microspheres at the point of manufacture in ordered to fulfil their intended function before transportation. Bulk emulsions are sensitized at the point of use on-sites. The difference in chemical properties between these two categories of emulsion is hence mainly due to the timing of presence of sensitizer.

12.6.2.9 Matrix or bulk emulsion (no voids) is not shock-sensitive since there is no known mechanism for the shock front to propagate. Also, heating a void-free liquid requires a very high pressure.

12.6.2.10 A local stimulus generating ‘hot spots’, such as sparks, friction, impact, static electricity, extreme ambient temperature, does not cause packaged emulsions (sensitized) to readily deflagrate under normal atmospheric conditions. To generate a deflagration which may subsequently transit to a detonation, a pressure in excess of 5 bars above atmospheric pressure is additionally required in the “deflagration mass”.

12.6.2.11 The behaviour of packaged emulsion following a shock or thermal stimulus is discussed in the following sections.

12.6.3 Accidental Packed Emulsion Initiation by Fire

12.6.3.1 Pools of molten AN may be formed in a fire. They may explode particularly if they are contaminated with other materials such as copper. AN may also melt and decompose with a release of toxic fumes which are mainly oxides of nitrogen. AN’s sensitivity to local stimuli increases when the temperature is beyond 140˚C [3] or in its molten form.

12.6.3.2 When the explosives are subjected to fire engulfment, many of them ignite and burn, deflagrate, and even detonate in some cases. These were indicated by a number of tests. The time for an explosive to ignite is dependent on its physical characteristics and chemical composition.

12.6.3.3 Cartridged emulsions are generally considered less sensitive to fire engulfment as a mean of initiation due to their high water content. However, the water content of the emulsion will be driven off when exposed to heat or fire. If the energy level of the heat is high enough, of long duration and with confinement pressure increases, cartridge emulsions may initiate.

12.6.3.4 The temperature of any reactive medium will be clearly raised by a fire surrounding the explosive load and enable evaporation of components (e.g. water). The rate at which evaporation occurs is dependent on the extent of fire and the heat transfer based on the design of the cargo container wall. The external part of the container wall will be heated directly with the flame. Heat will eventually be transferred to the explosive loads.

12.6.3.5 The transport accident statistics for ANFO indicate a minimum time to deflagration is about 30 minutes. Emulsions are considered more difficult to be initiated than ANFO as they contain higher water content.

12.6.3.6 The consequences of an accidental explosion due to thermal stimulus could be a thermal explosion or detonation or sometimes a combination of the two.

12.6.4 Accidental Packed Emulsion Initiation by Means Other Than Fire

12.6.4.1 There are commonly two distinct groups of non-fire initiation mechanisms: mechanical and electrical energy. Both shock and friction initiation are classified as mechanical as in most accidental situations, they are difficult to be distinguished. It has been recorded that some non-emulsion type explosives can initiate (in the absence of piercing) mechanically at an impact velocity as low as 15 m/s. If the explosives are pierced, it is likely that the required velocity will be far less than 15m/s. It is because localised heat generation resulting from frictional rubbing between layers of explosives, and is regarded as ‘stab-initiation’.

12.6.4.2 However, as demonstrated by the bullet impact test from a high velocity projectile, cartridged emulsions are insensitive to initiation by impact. According to the bullet impact test, it requires at least 10 times the energy level of that required to detonate nitroglycerine (NG) based explosives.

12.6.4.3 There are minimum ignition energy levels for all explosives, above which initiation will occur. Minimum ignition energy levels typically range between 0.015J and 1.26J.

12.6.4.4 The required ignition energy level of most explosives, including cartridged emulsions, is far exceeded by contact with mains electricity. The energy levels possible from batteries or alternators fitted to motor vehicles, or that due to static build-up on clothing are typically less than that required to initiate most commercial explosives (e.g. 0.02J or less). Therefore, only very sensitive explosives are likely to be ignited from these electrical energy sources and electrical energy is not a possible mean of initiation for the types of explosives used in this Project.

12.6.4.5 Water loss and prolonged temperature cycling above and below 34°C are possible causes of degradation of cartridged emulsion. Degradation of cartridged emulsion leads to potential caking or a change in ammonium nitrate crystalline state and increase in volume. Detonation by means other than fire is not caused by both modes of degradation.

12.6.5 Hazard Properties of Detonating Devices

12.6.5.1 Detonating devices may detonate when exposed to heat or flame, or with friction, impact, heat, low-level electrical current or electrostatic energy. Detonation produces shrapnel. Hazardous gases or vapours produced in fire could be lead fumes, nitrogen oxides and carbon monoxide. Nevertheless, these gases depend on the type of material used in the detonators.

12.6.5.2 Pentaerythritol tetranitrate (PETN) is the main explosive component in detonating devices including detonating cords and detonators. A primary explosive substance (e.g. lead azide) is included in detonators as it is very sensitive to initiation.

12.6.5.3 PETN in detonating cord has similar sensitivities to NG based explosives and it is generally more sensitive than emulsions.

12.6.5.4 PETN has the potential to deflagrate at ambient pressure following a local stimulus. A deflagration under ambient pressure or higher can be led by a local initiation. A detonation may occur from a deflagration. As an explosive, it has a comparatively small critical diameter for detonation. PETN has a shock sensitivity higher than emulsions but lower than NG based explosives. According to the bullet impact test, it also requires at least 10 times the energy level of that required to detonate an NG based explosive [3].

12.6.6 Accidental Initiation Associated with Storage at Magazine

12.6.6.1 The possible means of accidental initiation of the explosives at the proposed magazine include the followings:

- Inadequately controlled maintenance work;
- Improper method of work;
- Poor housekeeping;
- Electric fault within the store;
- Arson;
- Dropping of explosives during handling (applicable to detonators only); and
- Crushing of explosives under the wheel of vehicles during loading and unloading (applicable to detonators and detonating cords only)

12.6.6.2 The detonators are packaged within plastic separating strips, and the initiation of a single detonator will not propagate to the adjacent detonator. The packaged detonators are classified as Class 1.4B explosives, and the total mass of detonators is negligible in terms of the total explosive mass in storage.

12.6.7 Accidental Initiation Associated with Transportation from Magazine

12.6.7.1 The cartridged emulsion and detonating cords will be transported together within the same compartment on a truck. The vehicle cargo is designed to minimise all sources of local stimulus, only a significant crash impact or a fire will cause a concern to the explosives. A low speed traffic accident is unlikely to cause a concern to the explosives as stated in the ACDS study [5]. As conservative approach is adopted in this study, low speed traffic accident is still considered possible but with a lower probability [3]. Based on the bullet tests and review with explosives specialists, the activation energy of PETN or emulsion explosives is one order of magnitude higher than nitroglycerine (NG). Therefore, the probability of imitation under impact conditions can be reduced by one order of magnitude based on impact energy consideration [3] since NG was considered as the basis in previous studies (assessed at 0.001).

12.6.7.2 Time and possibility to full fire development on the vehicle (typically 5-10 minutes) and the amount of heat transferred to the loads are the major leading causes to the response of the explosives to an accidental fire. For emulsion explosives, if they are isolated from detonating cords, it may take at least another 30 minutes for the explosives to reach critical conditions based on accident statistics. For mixed loads of cartridged emulsions and detonating cords, this time may be considerably lowered but no precise time can be predicted from detonating cord transport accident data [3].

12.6.7.3 The behaviour of explosives used in this project as transported was considered to be similar to the XRL Study [2]. In the XRL Study, a review was conducted on the explosive properties with assistance from specialists in the explosives industry. The main findings for emulsion based explosives are as follows.

12.6.7.4 The radical change in explosive properties at higher temperatures compared to the original emulsion must be taken into account. At high temperatures (> melting point), emulsion explosives would lose water content which may result in a refined explosive (small droplet/ crystal size Ammonium Nitrate (AN)). This could lead to a thermal explosion, deflagration or detonation and the probability of 0.1 may not therefore be applicable to emulsion. Also, some limited accident statistics have some bearing on this hazard scenario: these accidents may include a combination of both thermal and mechanical stimuli, which would likely have resulted in explosion or detonation. The consensus was that the probability of an explosion for the case of an emulsion was less than 0.5 but further refinement of this upper estimate would require additional data and more detailed analysis [2].

12.6.8 Hazards Associated with Blasting Process

12.6.8.1 Hazards from the blasting process are considered to be the hazards induced by the blasting of a blast face. The design of the blast face is determined by the permitted vibration level of the sensitive receivers which is expected not to cause any damages to the receivers. However, potential hazards may occur when the process is completed with deviation with the designed process. Higher than expected vibration may be induced by such events.

12.6.8.2 According to WIL Study [1], the major hazard from blasting operations is flying debris. Flying debris is identified as a rock that has been propelled beyond the blasting area by the force of an explosion. The cause of flying debris is mismatch

of the distribution of explosive energy, type of confinement of the explosive charge, mechanical strength of the rock and lack of security measures at blasting area.

12.6.8.3 In this project, the effect of overpressure and flying debris is not considered as a blast door is in place and closed during the blasting process.

12.6.9 Incident Review

12.6.9.1 Reported safety incidents involving explosives for commercial and industrial applications were reviewed. The incident records were extracted from the UK Health and Safety Executive (UK HSE)'s Explosives Incidents Databased Advisory Services (EIDAS), US Mine Safety and Health Administration (MHSA), Western Australia's Department of Consumer and Employment Protection (DOCEP) and Hong Kong SAR Government's Annual Controlling Officers Report.

Explosives Storage related Incidents

12.6.9.2 A UK study identified 79 major incidents related to manufacture and storage of explosives during the period from 1950 to 1997. A total of 16 major incidents were attributed to the storage of explosives, among which 13 incidents related to the storage of gunpowder, ammunition, nitroglycerine and fireworks, 1 incident related to the storage of detonators and the remaining 2 incidents related to the storage of blasting explosives.

12.6.9.3 From the above study, it can be seen that the protection of explosives from malicious human activity and the elimination of possible ignition sources are crucial for maintaining storage facilities.

12.6.9.4 Some identified initiating causes of accidents in storage facilities are listed below:

- Impact;
- Friction;
- Overheating;
- Electrical effects (lightning/static discharges);
- Sparks;
- Spontaneous reactions; and
- Malicious action/mishandling.

12.6.9.5 To avoid incidents happening in the storage area, the only actions we can do are maintaining good housekeeping practice, eliminating potential ignition sources and putting explosives into safe and secure storage space. Applicability of the above-mentioned causes to the explosives magazine used in this Project are further discussed in **Section 12.7**.

Explosives Transport related Incidents

12.6.9.6 The incident records from CEDD website and different news platform in Hong Kong have been reviewed up to Year 2024 and there was no incident record related to

road transport of explosives. However, a minor incident happened at Queens Road West which involved a Mines Division explosives delivery vehicle in September 2010. Fortuitously, the crash impact was not significant, so that the integrity of the explosives was not affected.

12.6.9.7 The EIDAS (Explosives Incidents Database Advisory Service) database obtained most of the worldwide incidents related to the transport of commercial explosives reported from 1950 to 2017. It has identified a transport incident related to emulsion in which a tyre fire on a truck has spread to the explosive load and detonated, producing a massive hole. This incident did not cause any fatality as the truck crew had time to leave the truck and maintain a safe distance before the tyre fire reached the emulsion load and exploded.

12.6.9.8 There were also some incidents involving mixed cargoes of emulsion or watergel carried with other types of explosives. The EIDAS database also found 2 fire incidents associating explosives delivery vehicles in Australia in 1998 and 2007 respectively and all of these incidents did not cause any injuries or fatalities.

Explosives Use related Incidents

12.6.9.9 Incidents from 2000 to 2017 obtained from the EIDAS database have also been reviewed and there were 3 incidents related to the use of explosives. The first incident took place in India in 2007, a worker was injured when the electric detonators he was carrying exploded and the worker had been contracted to blast rocks. The second incident happened in Russia in 2008, an explosion occurred during the pneumatic loading of an ANFO into underground holes and caused 13 fatalities and 5 injuries. The third incident occurred in 2012 and it caused 1 fatality due to an explosion occurred in the powder feeder of the shock tube extrusion line which has communicated back to the magazine and detonated a few seconds later.

12.6.9.10 There was a total of 54,694 blasts in Hong Kong in the period between 2005 and 2024, which included both underground and surface blasts. During this period, there was no incident related to underground blasting operations.

12.6.10 Hazardous Scenarios for Risk Assessment

Storage of Explosives in Magazine

12.6.10.1 The explosives will be stored in 2 stores at the magazine in Tai Shu Ha. Within each store, explosives and detonators will be stored in separate compartments. The stores are designed with separation and enclosed walls so that initiation of the contents of one store will not affect other stores.

12.6.10.2 Possible hazardous scenarios associated with the storage of explosives are:

- Accidental detonation of a full load of explosives stored within one store.

12.6.10.3 The explosives load considered are listed in **Table 12.9**. The detonator explosives load was considered in the total explosives load.

Table 12.9 Explosives Storage Quantities

Magazine	Mass of explosives per site (TNT eqv. kg)	No. of stores	TNT equivalent per store (TNT eqv. kg)
Tai Shu Ha	800	2	400

Remarks:
1. Assumed 40% detonating cord & 60% cartridged emulsion.
2. Detonating cords are made of PETN.
3. Each detonator contains about 0.9g PETN.
4. 1kg of cartridged emulsion equals 0.96kg of TNT, and 1kg PETN equals 1.4kg of TNT.

Transportation of Explosives

12.6.10.4 Possible hazardous scenarios associated with the transport of explosives are:

- Accidental detonation of a full load of explosives on a Contractor's explosives delivery vehicle during the transfer from explosives magazine to tunnel portals; and

- Accidental detonation of a full load of explosives on a Contractor's explosives delivery vehicle during the transfer from tunnel portals to the blast faces.

12.6.10.5 Cartridged emulsions, detonators and detonating cords are transported by a licensed diesel vehicle from the explosives magazine to the project area and then to the blast faces. Potential hazards include higher than expected vibration and air overpressure due to detonation of full load of explosives during transportation.

12.6.10.6 A loading of 60 TNT eqv. kg to 200 TNT eqv. kg for the Contractor's explosive delivery trucks were assumed under this QRA.

Use of Explosives

12.6.10.7 Hazards from the blasting process are considered to be the hazards induced by the blasting of a blast face. The design of the blast face is determined by the permitted vibration level of the sensitive receivers which is expected not to cause any damages to the receivers. However, potential hazards may occur when the process is completed with deviation with the designed process. Higher than expected vibration may be induced by such events.

12.6.10.8 Possible hazardous scenarios associated with the use of explosives are:

- Higher than expected vibration generated by the blast faces during the blasting process.

12.6.10.9 According to WIL Study [1], the major hazard from blasting operations is flying debris. Flying debris is identified as a rock that has been propelled beyond the blasting area by the force of an explosion. The cause of flying debris is mismatch of the distribution of explosive energy, type of confinement of the explosive charge, mechanical strength of the rock and lack of security measures at blasting area.

12.6.10.10 In this QRA, the effects of overpressure and flying debris during tunnel blasting were not considered as a blast door will be in place and fully closed during the blasting process.

Ground Vibration Associated with the Use of Explosives

12.6.10.11 During rock excavation, ground vibration is a potential hazard if the stress wave intensity is high enough to induce a high level vibration. It is observed that peak particle velocity (PPV) is a good indicator for estimating the damage to structures. In general, reinforced concrete structures in good condition can readily tolerate a peak particle velocity of 50 mm/s without any risk of damage [15]. However, it is possible that there will be amplification between the peak ground motion and that experienced by the structure, therefore a peak particle velocity of the ground motion of 25 mm/s is used locally and internationally to prevent damage to buildings. This criterion is specified, for example, by the MTR for its structures when subjected to transient vibrations and also by the Hong Kong SAR General Specification for Civil Engineering Works.

12.6.10.12 Ground vibrations induced by this stress wave have a peak velocity that is related to the Maximum Instantaneous Charge (MIC) and the distance from the blast source. **Diagram 12-5** presents the typical range of charge weights and predicted vibration levels using the Mines Department vibration constants.

Radial Distance from Blast (m)	Maximum Instantenuous Charge (MIC) (kg)								
	10	8	6	5	4	2	1	0.5	0.25
	Predicted PPV (mm/s)								
160	5.37	4.69	3.93	3.52	3.07	2.01	1.32	0.86	0.57
140	6.32	5.51	4.63	4.14	3.61	2.37	1.55	1.02	0.67
120	7.63	6.66	5.58	5.00	4.36	2.86	1.87	1.23	0.80
100	9.53	8.31	6.98	6.24	5.45	3.57	2.34	1.53	1.00
90	10.83	9.45	7.93	7.10	6.19	4.06	2.66	1.74	1.14
80	12.51	10.91	9.16	8.19	7.15	4.69	3.07	2.01	1.32
70	14.70	12.85	10.78	9.64	8.42	5.51	3.61	2.37	1.55
60	17.80	15.50	13.00	11.64	10.16	6.66	4.36	2.86	1.87
50	22.20	19.40	16.20	14.50	12.69	8.31	5.45	3.57	2.34
40	29.13	25.42	21.33	19.09	16.66	10.91	7.15	4.69	3.07
30	41.38	36.11	30.30	27.11	23.66	15.50	10.16	6.66	4.36
20	67.86	59.23	49.69	44.46	38.80	25.42	16.70	10.91	7.15

5mm/s
13mm/s
25mm/s
>25mm/s

Diagram 12-5 Maximum Instantaneous Charge (MIC) versus Distance and Predicted Peak Particle Velocity (PPV)

12.6.10.13 According to **Diagram 12-5**, it is considered that structures in vicinity to the blasting site are unlikely to be subjected to PPV levels greater than 100mm/s or 229mm/s for object falling and structural damage for normal blasting operations.

12.6.10.14 Geo Guide 4 [7] identifies that there can be various levels of damage considered for structures based upon previous records for ground vibrations and ground frequencies. It also identifies that the potential for damage to a structure can be related to the natural frequency of the structure especially if the ground vibration has a similar characteristic frequency to the natural frequency of the building.

12.6.10.15 In general, it is considered that for buildings the formula to determine the approximate natural frequency can be approximated to the following:

Natural Building Frequency = 46 / height of the building (m)

12.6.10.16 If the ground motion frequency is similar to the building natural frequency, amplification could occur leading to larger motion of the building. This usually occurs when low frequency ground motion occurs over a long period of time such as in earthquakes that last maybe 30 seconds to minutes, rather than the usual 4 to 9 seconds for a tunnel blast.

12.6.10.17 Therefore, it is generally considered that buildings can readily tolerate a peak particle velocity of 100 mm/s for any falling objects that cause 1% fatality. Also, buildings are generally considered to have structural damages when a PPV of 229 mm/s is experienced. This criterion is generally used in previous similar studies like WIL Study [1].

12.7 Frequency Analysis

12.7.1.1 The frequency assessment consists of three parts. The occurrence frequency of accidental detonation of explosives during storage of explosives within magazine is the first part; while the occurrence frequency of accidental detonation of explosives during transportation from magazine to blasting sites is the second part; and the occurrence frequency of higher than expected ground vibration due to errors in the blasting process are the third part.

12.7.2 Frequency of Accidental Detonation of Explosives stored in Magazine Site

12.7.2.1 Explosives stored in the magazine could be initiated by the following causes:

- Generic causes
- Manual transfer from store to contractor's explosives delivery vehicles
- Lightning strike
- Aircraft Crash
- Earthquake
- Hill / vegetation fire
- Escalation

Generic Causes

12.7.2.2 The generic causes of all explosions in UK magazines (other than military stores and ordnance factories) were unstable explosive material caused by product

degradation, corrosion, and contamination; escalation of an external incident such as fire; or malicious acts such as vandalism or attempted theft. A generic failure frequency of 1×10^{-4} per year per magazine site were adopted in previous similar QRA studies [2][4][15].

12.7.2.3 All detonators will be stored in a dedicated chamber in the magazine. The explosives stored in the magazine will be protected from external fire since they will be housed inside a concrete or brick wall building, and the provision of fire-fighting measures could further lower the probability of initiation due to external fire.

12.7.2.4 The magazine will be provided with a comprehensive security system to reduce the possibility of vandalism or robbery. With provision of the above measures, the failure rate of 1×10^{-4} per year per magazine site is considered conservative and retained to represent all generic causes of explosion that are comment to nearly all magazines. Other site specified causes are addressed separated in following sections.

Manual Transfer from Store to Contractor's Explosives Delivery Vehicles

12.7.2.5 Explosives will be transferred from the explosives store to the explosives delivery vehicle or vice versa manually without the use of any tools which are susceptible to initiate the explosives. Failure due to manual transfer is already covered in the generic failure frequency mentioned above and was not assessed separately.

Lightning Strike

12.7.2.6 The explosive magazine will be a ground facility provided with lightning conductors to safeguard the facilities from direct lightning strikes and the grounding will be inspected regularly. The lightning protection installations should follow the standards BS EN 62305 or equivalent. No additional risk due to lightning strike compared to the UK magazines is anticipated. Failure due to lightning strike is already covered in the generic failure frequency and was not assessed separately.

Aircraft Crash

12.7.2.7 Aircrafts crashing into the magazine will be taken into account in this QRA by using the methodology given in HSE (1997) [28] for calculation of aircraft crash frequency. Calculation of aircraft crash frequency is provided in **Appendix 12.6**. Since the calculated failure rate is much smaller than order of 10^{-9}, failure caused by aircraft crash was not further considered in this assessment.

Earthquake

12.7.2.8 Hong Kong is a region of low seismicity [22][23], it is is not at disproportionate risk from earthquakes compared to similar explosives magazines worldwide (Hong Kong is approximate 600km away from the nearest Circum-Pacific Seismic Belt which runs through Japan, Taiwan and the Philippines). The generic failure frequency adopted is based on historical incidents with earthquakes already included in their cause of failure; it is considered that it is not necessary to address the failure due to earthquake separately.

Hill / Vegetation Fire

12.7.2.9 Hill / vegetation fires are quite common in Hong Kong, and the magazine could be potentially affected. According to the statistics data in the Annual Report published by the Agriculture, Fisheries and Conservation Department, there are 11-29 hill fires per year between 2014-2015 and 2023-2024, and the average vegetation area affected by fire was around 0.55% each year (**Table 12.10**), frequency of hill / vegetation affecting a specific site is estimated to be 5.5×10^{-3} per year.

Table 12.10 Hill Fire Data for Hong Kong

Year	Number of Hill Fire	Area Affected (Ha)	% of Total Country Park Affected
2014-2015	15	97	0.22
2015-2016	11	147	0.33
2016-2017	20	306	0.69
2017-2018	25	106	0.24
2018-2019	17	197	0.44
2019-2020	24	425	0.96
2020-2021	29	660	1.49
2021-2022	21	49	0.11
2022-2023	24	373	0.84
2023-2024	18	79	0.18

12.7.2.10 The explosive magazine is to be constructed of fire resistance materials such as bricks, cement rendering and steel doors, and the ground surface is to be constructed of concrete or stone to prevent fire ingress to the explosive store. Moreover, the land within the magazine site will be cleared of vegetation to remove any combustible materials, and fire-fighting measures will be in place. With consideration of the above, the chance of explosives being initiated due to hill / vegetation fire is considered to be negligible. Failure due to hill fire is already covered in the generic failure frequency mentioned above and was not assessed separately.

Escalation

12.7.2.11 An Ardeer Double Cartridge (ADC) test for cartridged emulsion showed that the consequence of a detonation is not able to propagate once the separation distance is beyond 2 cartridge diameters. Previous study [4] considered that it is impossible for an explosion within one magazine store to directly initiate an explosion within an adjacent store, hence the direct propagation by blast pressure wave and thermal radiation effects of an explosion within one store initiating an explosion with an adjacent store is not considered. However, ground shock induced from an explosion may cause damage within the adjacent stores if the vibration level exceeds the vibration threshold of the store structure, and leading to subsequent explosion.

12.7.2.12 Ground vibrations can be assessed by the following equation,

$$A = K(R/Q^d)^{-b}$$

where

A = predicted particle velocity in mm/s
K = a 'rock constant'
Q = mass of explosives detonated of each storage = 400 TNT eqv. kg
R = distance in meters between the blast and the measuring point
d = charge exponent, assumed to be 0.5 [6]
b = attenuation exponent, assumed to be 1.22 [3]

12.7.2.13 A building can withstand a vibration level lower than 229 mm/s without significant structural damage. From the International Society of Explosives Engineers (ISEE) handbook [24], a range of rock constant K = 173 to K = 4320 is identified for construction activities, depending on the degree of confinement. The rock constant K for aboveground storage of explosives is hence conservatively considered as 200 since there is no coupling with the ground.

12.7.2.14 The separation distance between 2 adjacent stores is 18m. With this separation distance, the maximum ground vibration at a store will be 227mm/s when a full load of 400 TNT eqv. kg of explosives is initiated at the adjacent store. Since the maximum ground vibration will be less than 229 mm/s, the possibility of explosives within adjacent stores being initiated is considered negligible.

12.7.3 Frequency of Accidental Detonation of Explosives during Transportation

12.7.3.1 Explosives during transport from magazine site to the blasting sites could be initiated by the following causes:

- Non-crash fire;
- Crash fire;
- Crash impact; and
- Spontaneous explosion of 'unsafe explosives'.

12.7.3.2 For non-crash fire and crash fire, the explosives load will be initiated once the load is engulfed by a fire for a period of time.

12.7.3.3 For crash impact, a significant mechanical impact during vehicle collision is required to affect the stability of the explosives load and initiate the explosion.

12.7.3.4 For spontaneous explosion, it is mainly due to badly packaged or manufactured explosives, and / or explosives which do not meet the specifications.

12.7.3.5 The ACDS study [5] assessed risks related to the transport of explosives in ports. The DNV study [8] then adjust the basic frequencies presented in ACDS to address the risk associated with transport of commercial explosives by Mines Division explosives delivery vehicles. Previous similar studies like the SIL (East) Study [4], XRL Study [2], SCL Study [3], WIL Study [1] and Shatin Cavern Study [15], all

adopted the frequencies derived in DNV study for the transport of explosives in trucks operated by contractors from explosives magazine to construction-sites, and fine-tuned the failure frequencies based on the latest knowledge on the explosives' properties, vehicle impact frequencies and specific design features of delivery trucks. Derivations of each frequency component are presented in the XRL Study [2]. Some minor modifications for the frequencies were made, taking into account the latest vehicle impact frequencies in Hong Kong, to calculate the failure frequency of contractor's explosives delivery vehicles; whereas the transport risk of Mines Division's explosives delivery vehicles is under Mines Division's purview and falls outside the scope of this EIA.

Historical Light Goods Vehicles (LGVs) and Medium & Heavy Goods Vehicles (MGVs & HGVs) Accidents in Hong Kong

12.7.3.6 A review of the Hong Kong accident data for LGVs and MGVs & HGVs was carried out based on the latest Annual Road Traffic Accident Statistics data [16] published by the Transport Department. The overall number of accident involvements per kilometer is given in **Table 12.11** for LGVs and MGVs & HGVs and all motor vehicles. Using a five-year average, the LGVs and MGVs & HGVs involvement per million vehicle kilometer (pmvkm) are 1.33 and 1.02 involvements pmvkm respectively.

Table 12.11 Hong Kong Vehicle Accident Involvements by Vehicle Type

Vehicle involvements	2020	2021	2022	2023	2024	5 years average
Light Goods Vehicles						
No. involved in accident	2709	3316	2956	3248	3469	3140
% of total involvement	12.3%	12.7%	13.1%	12.2%	11.9%	12.4%
No. licensed (mid-year)	73293	74879	75855	75551	73220	74560
Annual mvkm	2281	2420	2466	2390	2283	2368
Involvement rate: per 1,000 vehicles	37.0	44.3	39.0	43.0	47.4	42.1
Involvement rate: pmvkm	1.19	1.37	1.20	1.36	1.52	1.33
Medium & Heavy Goods Vehicles						
No. involved in accident	1008	1204	1048	1073	1261	1119
% of total involvement	4.6%	4.6%	4.7%	4.0%	4.3%	4.4%
No. licensed (mid-year)	42150	41721	40954	40817	40113	41151
Annual mvkm	1105	1179	1105	1065	1025	1095.8
Involvement rate: per 1,000 vehicles	23.9	28.9	25.6	26.3	31.4	27.2
Involvement rate: pmvkm	0.91	1.02	0.95	1.01	1.23	1.02
Total						
No. involved in accident	22063	26190	22499	26614	29239	25321
No. licensed (mid-year)	797262	815203	810413	819129	811084	810618
Annual mvkm	13344	14143	14018	14662	14532	14140
Involvement rate: per 1,000 vehicles	27.7	32.1	27.8	32.5	36.0	31.21
Involvement rate: pmvkm	1.65	1.85	1.61	1.82	2.01	1.79

Severe Impact Accident Involvements for LGVs and MGVs & HGVs in Hong Kong

12.7.3.7 A review of fatal and serious accidents is shown in **Table 12.12**. Accounting for only fatal and serious accidents, the LGVs and MGVs & HGVs significant accident involvement rate is estimated to be around at 0.08 pmvkm and 0.10 pmvkm respectively.

Table 12.12 Hong Kong Vehicle Accident Involvements by Severity

Serious and Fatal Vehicle involvements	2020	2021	2022	2023	2024	5 years average
Involvement rate: per million veh-km						
LGVs	1.19	1.37	1.20	1.36	1.52	1.33
MGVs & HGVs	0.91	1.02	0.95	1.01	1.23	1.02
Total involvements						
LGVs	2709	3316	2956	3248	3469	3140
MGVs & HGVs	1008	1204	1048	1073	1261	1118.8
Fatal Involvements						
LGVs	14	20	28	17	16	19
MGVs & HGVs	17	21	17	13	14	16
Serious injury involvements						
LGVs	239	226	180	135	126	181
MGVs & HGVs	120	109	97	90	61	95
Slightly injury involvements						
LGVs	2897	3698	3330	3787	4176	3578
MGVs & HGVs	1065	1355	1145	1418	1506	1298
High impact accident involvement rate per million vehicle km						
LGVs	0.0061	0.0083	0.0114	0.0071	0.0070	0.0080
MGVs & HGVs	0.0154	0.0178	0.0154	0.0122	0.0137	0.0149
Medium impact accident involvement rate per million vehicle km						
LGVs	0.1048	0.0934	0.0730	0.0565	0.0552	0.0766
MGVs & HGVs	0.1086	0.0925	0.0878	0.0845	0.0595	0.0866
Minor impact accident involvement rate per million vehicle km						
LGVs	1.270	1.528	1.350	1.585	1.829	1.5124
MGVs & HGVs	0.964	1.149	1.036	1.331	1.469	1.1900

Historical Expressway and Non-Expressway Accidents in Hong Kong

12.7.3.8 The accident statistics for both expressway and non-expressway could be derived from the latest Annual Road Traffic Accident Statistics data [16]. Adjustment factors are derived based on fatal and serious accidents as shown in **Table 12.13**.

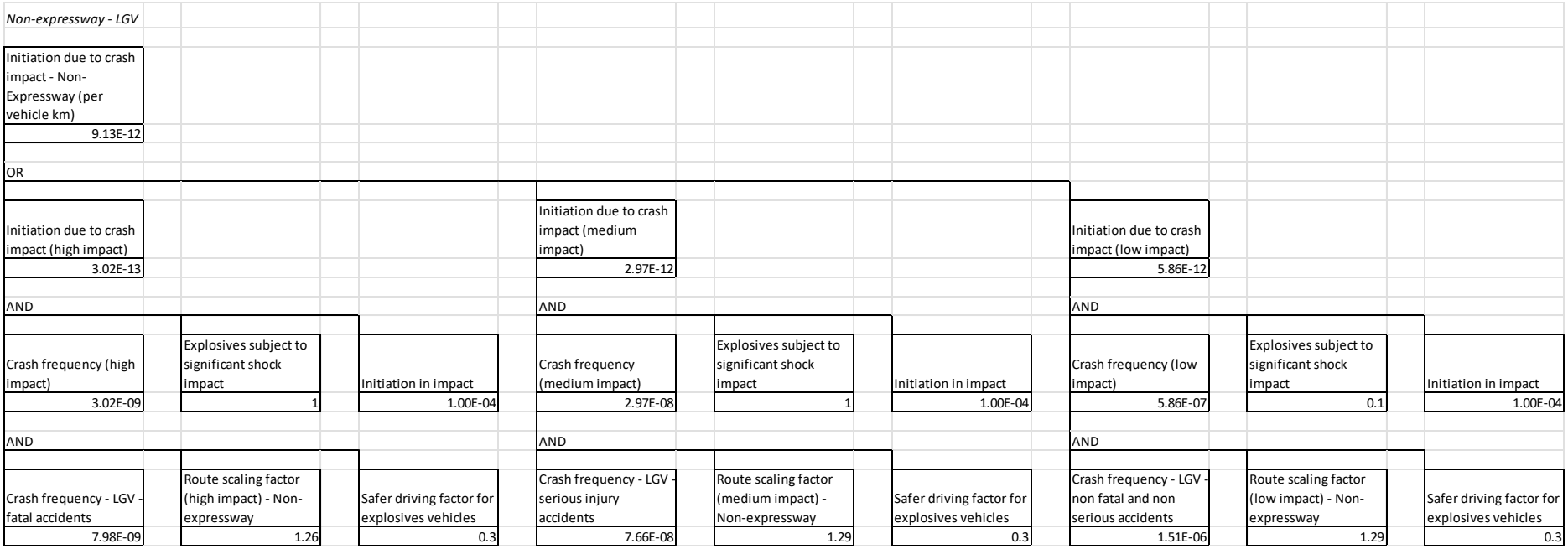
Table 12.13 Expressway and Non-expressway Involvement Rate (All Vehicle Types)

Rate of significant vehicle accidents per road type (all vehicles)	All expressways	Non-expressways	All roads
Accident rate pmvkm	0.38	1.63	1.26
Annual veh-km (in millions)	4276.3	10290.4	14566.7
Total Accidents	1625	16729	18354
Fatal Accidents	9	74	83
Serious Accidents	66	685	751
Fatal Accident Rate pmvkm	0.002	0.007	0.006
Serious Injury Accident Rate pmvkm	0.015	0.067	0.052
Fatal Accident Rate Ratio (compared to all roads as base case)	0.37	1.26	1
Serious Accident Rate (compared to all roads as base case)	0.30	1.29	1

Severe Impact Frequency with Potential to Damage the Explosives

12.7.3.9 The fault trees developed for non-expressway vehicle involvement scenario under the XRL study [2] were updated with the latest vehicle impact frequencies. The fault trees for contractor's explosives delivery vehicles is shown in **Diagram 12-6** and **Diagram 12-7**. The frequency of explosive load initiation due to impact and also the severe impact rates derived for the explosive delivery trucks are also shown in these diagrams.

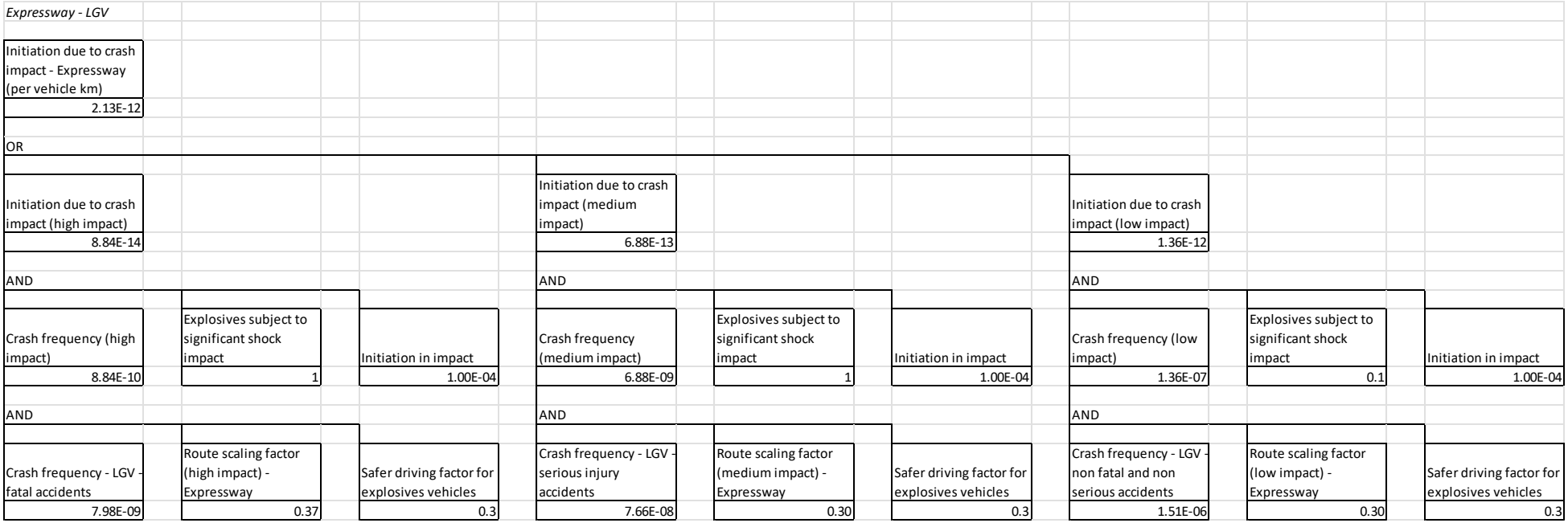
12.7.3.10 The total crash frequency for explosives delivery vehicles is 6.19×10^{-7} per vehicle-km on non-expressway and 1.44×10^{-7} per vehicle-km on expressway. Frequency of initiation of explosives on the explosives delivery vehicles due to crash impact was assessed to be 9.13×10^{-12} per vehicle-km on non-expressway and 2.13×10^{-12} per vehicle-km on expressway.



Note:

1. The total crash frequency for explosive vehicles is **6.19×10^{-7}** per vehicle-km (i.e. summation of crash frequencies of low, medium and high impact.

Diagram 12-6 Fault Tree for Explosives Load Initiation Due to Impact (LGV on Non-Expressway)



Note:

1. The total crash frequency for explosive vehicles is **1.44×10^{-7}** per vehicle-km (i.e. summation of crash frequencies of low, medium and high impact.

Diagram 12-7 Fault Tree for Explosives Load Initiation Due to Impact (LGV on Expressway)

Initiation due to Crash Fire

12.7.3.11 A crash fire explosive cargo damaging rate of 2.64×10^{-10} per vehicle-km reported by Moreton (1993) was quoted in the XRL EIA [2] with the UK reportable HGV vehicle accident involvement rate of 0.62 pmvkm in 1992. A conversion factor was applied to convert the UK vehicle involvement rate to HK involvement rate by considering the latest LGV and MGV involvement rates in 2023 in Hong Kong to UK involvement rate (0.62 pmvkm). The crash fire frequencies with explosives involved in fire on non-expressway and expressway in Hong Kong are 2.63×10^{-10} per vehicle-km and 6.11×10^{-10} per vehicle-km respectively, and the initiation of explosives on contractor's explosives delivery vehicles due to crash fire was assessed to be 1.32×10^{-10} per vehicle-km on non-expressway and 3.06×10^{-11} per vehicle-km on expressway.

Explosion Frequency Fault Tree

12.7.3.12 The explosives initiation fault tree inputs are presented in **Table 12.14** and the fault tree models for the road transport explosion are shown in **Diagram 12-8** for contractor's explosives delivery vehicles.

Table 12.14 Explosives Initiation Fault Tree Inputs

Event	Event Type	Value	Reference
Vehicle crash (on non-expressway)	Frequency	6.19×10^{-7} / vehicle km	Section 12.7.3.10
Vehicle crash (on expressway)	Frequency	1.44×10^{-7} / vehicle km	Section 12.7.3.10
Crash fire (on non-expressway)	Frequency	2.63×10^{-10} / vehicle km	Section 12.7.3.11
Crash fire (on expressway)	Frequency	6.11×10^{-11} / vehicle km	Section 12.7.3.11
Non-crash fire	Frequency	1.30×10^{-9} / vehicle km	Reference [2]
Explosives initiation in fire	Probability	0.5	Reference [2]
Explosives initiation in impact	Probability	0.0001	Reference [2]
Unsafe explosives	Probability	1% of overall frequency (initiation due to crash fire, non-crash fire and crash impact)	Reference [2]



Diagram 12-8 Explosives Initiation Fault Tree for Contractor's Explosives Delivery Vehicles

12.7.3.13 For contractor's explosive delivery vehicles, the overall frequency of accidental initiation during transportation of explosives is 7.97×10^{-10} per truck-km per year on non-expressway and 6.88×10^{-10} per truck-km per year on expressway.

12.7.3.14 Since the blasting will proceed from both portals of a tunnel and the transport length within the tunnels to be constructed under the Project varies as the blasting proceeds, the average transport length from each portal was assumed as one-fourth of the total length of the assessed tunnel for all deliveries. The calculated

frequency of accidental detonator of explosives during transport is shown in **Appendix 12.4**.

12.7.4 Frequency of Higher than Expected Ground Vibration due to Errors in the Blasting Process

12.7.4.1 The main causes of errors in the blasting process are due to human errors, which include errors in design manufacturing, installation, checking and recovery.

12.7.4.2 Human Error Assessment and Reduction Technique (HEART) is a human reliability assessment method based on human performance literature. It quantifies human error probabilities by assessing the interactions between humans, their specific tasks, performance shaping / human factors and error producing conditions. Review of the HEART analysis of WIL Study [1], Shatin Cavern Study [15] and Tsing Yi-Lantau Link Study [34] was carried out and the HEART analysis is presented in **Appendix 12.5**.

12.7.4.3 With reference to Tsing Yi-Lantau Link Study [34], the QRA considered failure scenarios of higher than expected vibration due to detonation of up to 6MIC at the same time. The fault tree analysis of accidental simultaneous detonation of 2MIC to 6MIC is shown in **Appendix 12.5**.

12.7.4.4 The probability of 2MIC to 6 MIC detonated at the same time per blast is summarised in **Table 12.15**. It was assumed that the occurrence probability of repeated error for either design or manufacturing of detonator is 0.01 (details please refer to **Appendix 12.5**). it can be observed that the occurrence probability for each additional MIC detonated at the same time will hence be roughly 2 orders of magnitude lower each time, the occurrence probability for 6MIC is at the order of magnitude of 10^{-14} per blast, or equivalent to the order of magnitude of 10^{-10} for Tunnel A with number of blast more than 1,200 per year. Due to the very low occurrence probability of 6MIC detonated at the same time, it was not further considered in the remaining sections of the risk assessment.

12.7.4.5 The number of blasts assumed for Tunnels A, B, C and E was 1,272, 760, 100 and 120 per year respectively. No concurrent blasting at Tunnel E (from both East Portal and West Portal) and Tunnel B (from South Portal) will be carried out, and also no concurrent blasting at Tunnel C (from both South Portal and North Portal) and Tunnel B (from North portal) will be carried out.

12.7.4.6 The risk assessment for the use of explosives was carried out every 10m chainage interval for the construction of tunnels.

12.7.4.7 The overall frequencies of failure scenarios leading to higher than expected vibration for this project are estimated accordingly and the overall frequencies are summarised in **Table 12.15**.

Table 12.15 Overall Frequencies of Failure Scenarios Leading to Higher than Expected Vibration During Tunnel Blasting

Sections	No. of blast per year	Occurrence Frequency for multiple MIC detonated at the same time per Section (Occurrence per Year)				
		2MIC	3MIC	4MIC	5MIC	6MIC [Note 1]
Per blast	--	1.12E-05	9.51E-08	9.51E-10	9.51E-12	9.51E-14
Tunnel A	1,272	1.43E-02	1.21E-04	1.21E-06	1.21E-08	1.21E-10
Tunnel B	760	8.54E-03	7.23E-05	7.23E-07	7.23E-09	7.23E-11
Tunnel C	100	1.12E-03	9.51E-06	9.51E-08	9.51E-10	9.51E-12
Tunnel E	120	1.35E-03	1.14E-05	1.14E-07	1.14E-09	1.14E-11

Note 1: Due to the very low occurrence probability of 6 MIC, it will not be further assessed in the remaining sections.

12.7.5 Domino Effect of Underground High Pressure Town Gas Pipeline

12.7.5.1 As mentioned in **Section 12.5.5.5**, there is an interfacing section of around 9.5km between the HP pipeline and the explosives transport route (i.e. from magazine in Tai Shu Ha to Explosives Unloading Point III on Kwu Tung Road), as shown in **Diagram 12-9**. The average travel speed of the explosives delivery vehicle along the whole transport route is about 50km/hr. Time fraction of an explosives delivery vehicle within the interface section is about 2.28×10^{-6} per km per year.

12.7.5.2 According to Electrical and Mechanical Services Department (EMSD)'s Guidance Note on Quantitative Risk Assessment Study for High Pressure Town Gas Installations in Hong Kong (HP Guidance), the failure frequency of underground high pressure town gas transmission pipelines is 1×10^{-5} per km per year. The event outcome of fireball or jet fire due to failure of the HK pipeline could trigger failure of explosives on the explosives delivery vehicles if the transport route is close to the HP pipeline. The event outcome frequency of fireball and jet fire was estimated to be 1.73×10^{-6} per km per year in the recently approved Route 11 EIA [19]. The domino effect of the HP pipeline would be about 3.94×10^{-12} per km per year, which is about 2 orders lower than accidental explosion during transport as discussed in **Section 12.7.3.13**. Moreover, a minimum cover of 1.1m is provided for the underground high pressure town gas transmission pipelines, with consideration of the presence factor and the protection by the earth cover, it is presumed that probability of failure of the HP pipeline triggering the failure of explosives during transportation is very unlikely and this was not further considered in the QRA.

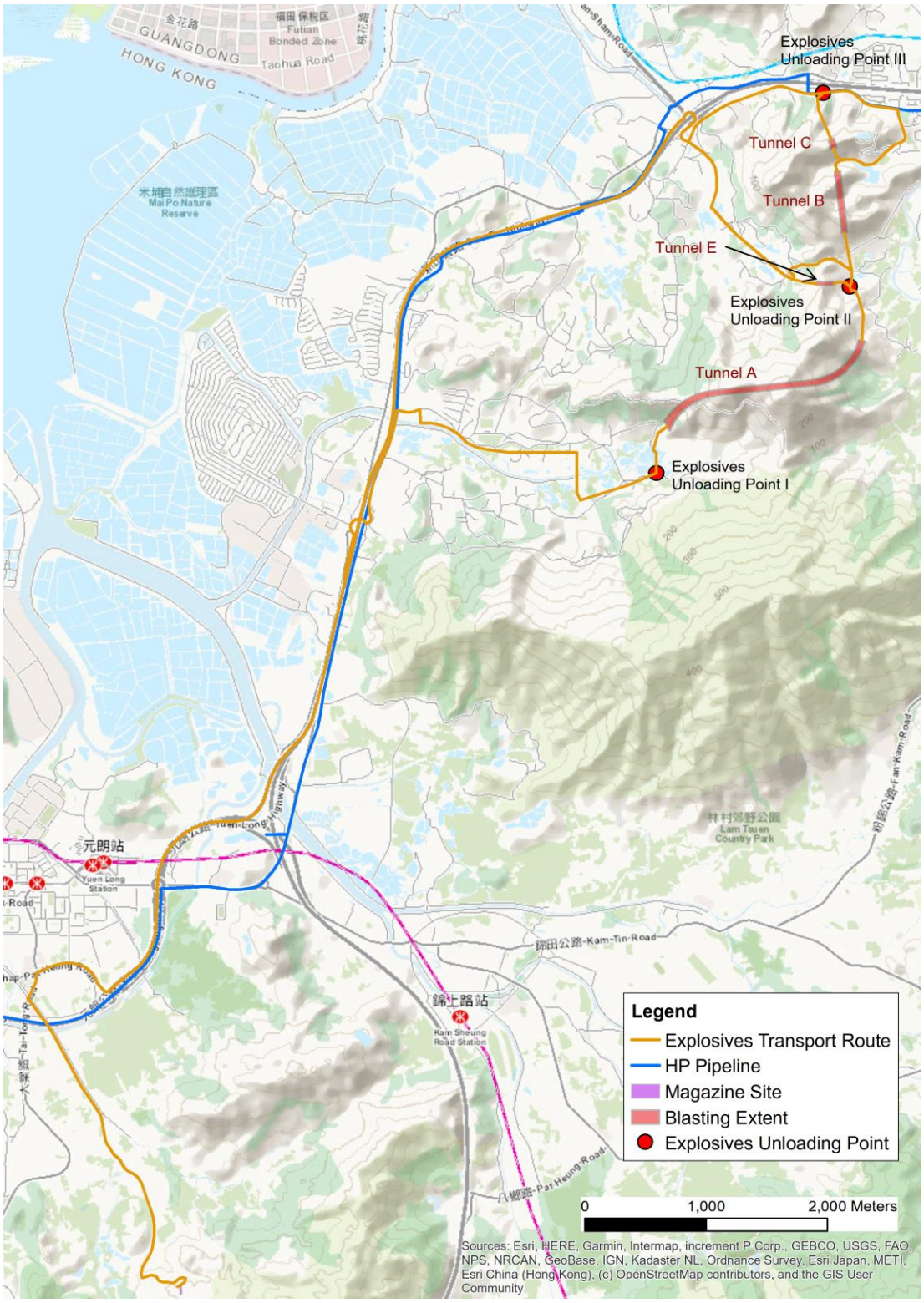


Diagram 12-9 Alignment of HP Pipeline overlapping with Transport Route

12.8 Consequence Analysis

12.8.1.1 Possible outcomes from hazardous events associated with transport and use of explosives include:

- **Primary hazards**: Blast effects and ground shock / vibrations;
- **Secondary hazards**: Effects on buildings, slopes and other sensitive receivers; and
- **Tertiary hazards**: Landslide and boulders fall.

12.8.2 Primary Hazards

Blast and Overpressure Waves Generated due to Accidental Detonation during Storage at Magazine Site and Transport of Explosives to Tunnel Portals

12.8.2.1 The *Explosives Storage and Transport Committee (ESTC)* model developed by the UK Health and Safety Committee (HSC) [9] is utilised to determine the probability of fatality due to blast and overpressure waves. The consequences of accidental explosion during storage at magazine site and transport of explosives from magazine to tunnel portals was assessed by the above ESTC model. The ESTC model analyses the blast effects for people indoors and outdoors separately.

People Indoors

$$log_{10}P = 1.827 - 3.433log_{10}S - 0.853(log_{10}S)^2 + 0.356(log_{10}S)^3$$

for $3 < S < 55$

where

$S = R/Q^{1/3}$;
P is the probability of death;
R is the range in meters; and
Q is the explosive charge mass in kg (TNT equivalent mass).

People Outdoors

$$P = \frac{e^{(-5.785S+19.047)}}{100}$$

for $2.5 < S < 5.3$

12.8.2.2 Population in vehicles and buildings were considered to be indoors and the indoor consequence model was applied; pedestrians were considered as outdoor populations and the outdoor consequence model was applied.

Blast Effects including Air Overpressure and Flying Debris due to Accidental Detonation during Transport of Explosives within Tunnels

12.8.2.3 The blast and overpressure effects for detonation of cartridges during transport in the tunnel can be estimated by the DoD 6055.9-STD equation C9.7-16 [14] and ESTC model. Comparisons will be made between the two methods when assessing the likelihood of fatalities due to blast effects.

12.8.2.4 DoD 6055.9-STD equation C9.7-16 is as follow:

$$R = 220.191\ D_{HYD}\ ((W/V_E)^{0.5} / P_{so})^{1/1.4}$$

Where

R is the distance from the opening (m);
D_{HYD} is the effective hydraulic diameter that controls dynamic flow issuing from the opening (D_{HYD} = 4A/P where A is the cross-sectional area of the opening and P is the perimeter);
Pso is the overpressure at distance R (kPa);
W is the charge weight for the maximum credible event (kg); and
V_E is the total volume engulfed by the blast wavefront within the tunnel system at the time the wavefront arrives at the point of interest (m^3).

12.8.2.5 If the point of interest is off the centreline axis of the opening at an angle θ, the distance versus overpressure can be evaluated from equation C9.7-17 of DoD 6055.9-STD.

$$R(\theta) = R(\theta = 0) / (1+(\theta/ 56)^2)^{1/1.4}$$

Where

R(θ = 0) is the distance along the centreline axis.

12.8.2.6 Overpressure that is sufficient to cause a fatality is based on that resulting in lung haemorrhage in people located outside. In general, human body is capable of adapting to large changes in pressure which occur gradually. A probit equation based on data relating to death primarily from lung haemorrhage due to peak overpressure was derived by Fugelso, Weiner and Schiffman [27],

$$Pr = -77.1 + 6.91 \ln P^0$$

Where

P^0 is the peak overpressure generated by the blast (Pa).

12.8.2.7 The peak overpressures corresponding to a 1%, 3%, 10%, 50%, and 90% fatality level were calculated to be 103 kPa, 110 kPa, 120 kPa, 144 kPa and 174 kPa respectively.

12.8.2.8 The lengths of Tunnel A, Tunnel B, Tunnel C and Tunnel E are around 2000m, 550m, 200m and 200m, with a typical cross-sectional area of 160m^2 for Tunnels A, B and C and a typical cross-sectional area of 122m^2 for Tunnel E. According to DoD 6055.9-STD Table C9.T32, the distance from the opening is the longest when θ = 0. In this assessment, point of interest for overpressure generated by detonation of 200 TNT eqv. kg cartridges at all tunnels is assumed to be along the centreline axis of the opening to obtain a maximum distance. Since a tunnel will be constructed from both ends, it was assumed that each explosives delivery vehicle will only travel half of the total length of tunnel. Similar to the calculation of frequency of accidental initiation of explosives during transportation from portal to blast face, one-fourth of the total length of the tunnel (i.e. half of the length of the tunnel to be travelled by each explosives delivery vehicle) was taken as the point of interest.

12.8.2.9 The hazard distances for the detonation of 200 TNT eqv. kg explosives calculated by both equation C9.7-16 of DoD 6055.9-STD and ESTC model are tabulated in **Table 12.16**. It can be seen from the table that the hazard distances estimated by ESTC model is generally more conservative than equation C9.7-16 of DoD 6055.9-

STD. Therefore, consequence distances obtained from the ESTC models were used to assess the risk of transporting explosives within tunnels.

12.8.2.10 To be consistent with previous similar studies, during the construction of tunnel, an initiation of explosives during transport within the main access tunnel is considered as an explosion at the portal of the main access tunnel since no decay factor was considered for a blast wave propagating from the blast face to the portals.

Table 12.16 Summary of Hazard Distances for Peak Overpressure at Various Fatality Level (Detonation of 200kg TNT eqv. kg Explosives)

Tunnel Sections	Hazard Distances for Peak Overpressure at various Fatality Level (m)				
	90%	50%	10%	3%	1%
Calculated by equation C9.7-16 of DoD 6055.9-STD					
Tunnel A	9.3	10.6	12.1	12.9	13.5
Tunnel B	14.7	16.8	19.2	20.4	21.4
Tunnel C	21.1	24.1	27.5	29.3	30.7
Tunnel E	20.3	23.2	26.5	28.2	29.5
Calculated by ESTC indoor model					
All Tunnels	18.4	21.2	31.5	42.4	54.6

Ground Shock / Vibration Generated by Rock Excavation using Explosives

12.8.2.11 Ground shock / vibration generated by an explosion can be calculated by the following equation [6]:

$$A = K(R/Q^{d})^{-b}$$

Where

A = predicted particle velocity in mm/s
K = a 'rock constant'
Q = maximum charge weight per delay interval in kilograms
R = distance in meters between the blast and the measuring point
d = charge exponent, assumed to be 0.5
b = attenuation exponent, assumed to be 1.22

12.8.2.12 The value of K is dependent to rock type as well as structure and confinement of the blast operation. The typical value of K for the granitic and volcanic rocks in Hong Kong ranges from 1,000 to 1,200 for rock blasting according to Geoguide 4 [7]. In this QRA, the rock constant, K, was considered as 1,200 for the rock excavation using explosives.

Ground Shock / Vibration Generated due to Accidental Detonation during Storage and Transport of Explosives

12.8.2.13 In this Project, the explosives usage is carried out underground while the storage and transport of explosives from magazine site to tunnel portals are all aboveground. There are some slopes and buildings identified along the transport routes of explosives that a landslide or a boulder fall may be triggered by explosion on an explosives delivery vehicle. Aboveground explosion will result in a lower pressure wave as the explosives are less confined. The consequence is considered less

concerned compared to the hazards posed by the overpressure wave and debris generated by the explosion.

12.8.2.14 The methodology to access the ground shock due to storage and transport of explosives is the same as **Section 12.8.2.11** above. Instead, the value of K is considered to be 200 to represent the "decoupling" of explosives during transport in the tunnel [11].

12.8.2.15 Landslide or boulder fall may be triggered by the explosion on an explosives delivery vehicle if slopes along the transport route of explosives fall due to ground vibration. However, the findings from WIL EIA [1] suggested that any landslide or boulder fall would only impact the same area along that road that was already affected by the primary explosion consequences. Since no significant additionally fatality would occur, secondary hazards regarding slope and boulders were not further considered.

12.8.2.16 There are some natural terrains near to the proposed magazine site. It is possible that an explosion inside the storage of the magazine would trigger a landslide or a boulder fall. The Tai Shu Ha magazine was also proposed in the NOL EIA [27], and it suggested that the landslide or boulder fall event would only impact in the area that is close to the magazine site which was already affected by the primary explosion consequences. Therefore, no significant additional fatality would occur, so secondary hazard was also not further considered.

12.8.3 Secondary Hazards

Effect on Building

12.8.3.1 In Hong Kong, the maximum values of peak particle velocity normally accepted by a building to prevent cosmetic damage is 25 mm/s. For the purpose of this QRA, the peak particle velocity that induces significant structural damage and results in potential fatalities is also required.

12.8.3.2 The US Bureau of Mines (USBoM) Bulletin 656 Blasting vibrations and their effects on structure were reviewed. A literature reviewed in Bulletin 656 obtained the damage level of a building with different PPVs. These results are tabulated in **Table 12.17**.

Table 12.17 Damage Level of Buildings due to Ground Vibration

Damage Level	Peak Particle Velocity, mm/s
No noticeable damage	70
Fine cracking and fall of plaster	110
Cracking	160
Serious cracking	231

12.8.3.3 The US Department of Defense Standard DoD 6055.9-STD 2004 DOD Ammunition and explosives safety standards [14] was also be reviewed for the maximum particle velocity induced in the ground at the building site.

12.8.3.4 Criteria adopted for building risk assessment were summarised as follow [1] [14]:

- PPV = 229 mm/s – Building structural collapse threshold; and

- PPV = 100 mm/s – Object fall threshold.

12.8.3.5 Falling objects is expected when a peak particle velocity reaches the object fall threshold 100 mm/s. Buildings will collapse only when a peak particle velocity is significantly larger than the assumed threshold limit 229 mm/s. The above criteria are considered conservative.

Building Collapse Models for Explosion / Earthquake

12.8.3.6 Building damage vulnerability models for partial collapse / damage was adopted in previous similar studies such as WIL study [1]. The objects with the potential to fall are assumed to be 1 m^2 large.

12.8.3.7 The major causes of fatality due to partial building collapse are collapse of roofs, ceilings and walls. They are considered as the most serious types of falling objects causing more than 1 fatality. The fatality rates calculated from different models for partial collapse of a building vary from 0.01% to 1.5%.

12.8.3.8 A fatality rate of 1% was considered for fatalities resulting from falling objects when the PPV value received at the buildings reaches 100 mm/s. The fatality probabilities of buildings receiving PPV values within the range of 100 mm/s and 229 mm/s (i.e. 1% fatality and 100% fatality respectively) were interpolated.

Effect on Slopes

12.8.3.9 The vibration limits for registered geotechnical features are different for each individual feature. GEO TGN 28 [26] allows the use of prescribed allowable PPV of 25 mm/s for soil cut slopes that have been upgraded and are in good condition. With reference to TGN 28, any slope that falls into Consequence-to-Life (CTL) Categories 1 and 2 and meets current standards shall also be considered that an allowable PPV of 25 mm/s could be adopted. Furthermore, any slope within Consequence-to-Life category 3 can be assigned with an allowable PPV of 25 mm/s.

12.8.3.10 PPV of 25 mm/s is the standard and prescribed allowable PPV for the existing slopes. As the vibration limits for registered geotechnical features are different for each individual feature, 90 mm/s is adopted as the screening criteria for this project. For any slope for which it is proposed to adopt the use of the prescriptive PPV, visual inspection will be carried out to confirm there are no signs of distress or instability, or any other stability concerns.

12.8.3.11 The analysis of the effects of vibration on the stability of slopes is based on the guidelines detailed in GEO Report No. 15 [11]. The critical Peak Particle Velocity (PPVc) corresponding to the maximum vibration is calculated using the following equation:

$$PPVc = K_c g/(\omega Ka)$$

Where

$K_c g$ = the critical acceleration at which the slope has a factor of safety of 1.0 against failure,
g = the acceleration due to gravity (ms^{-2});
ω = the circular frequency of the ground motion (2πf); and
Ka = the magnification factor.

12.8.3.12 The value of ω is related to f, the frequency of vibration. The typical range of the frequency is about 30-100Hz for blasting. As recommended in GEO Report No. 15, a frequency of 30Hz was adopted for this QRA.

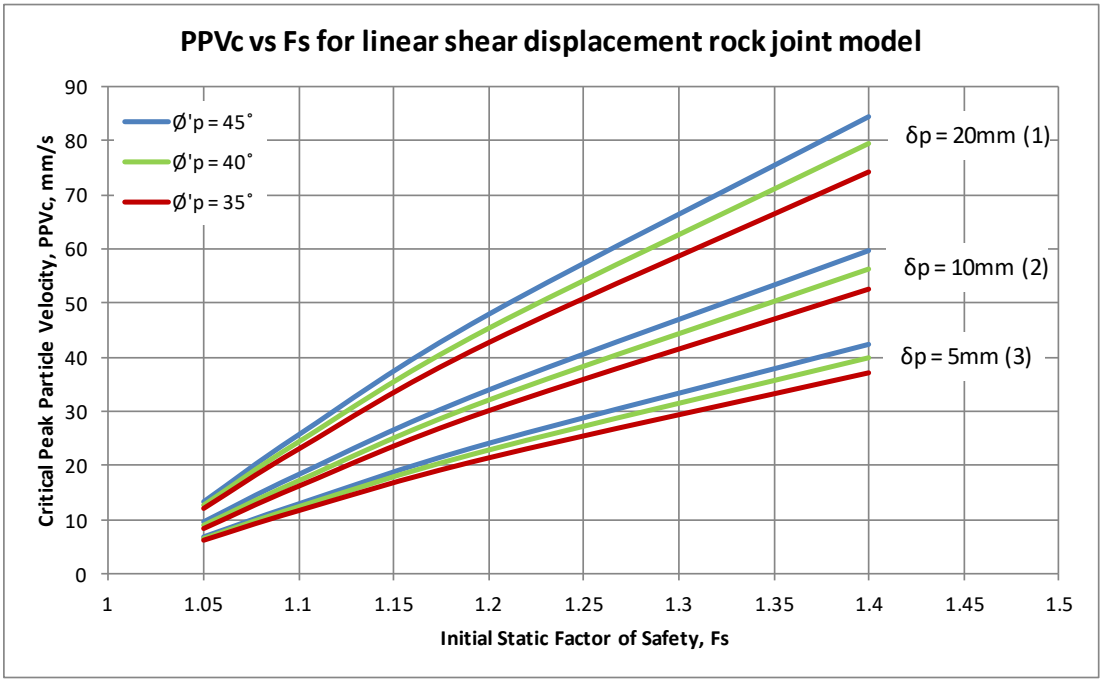
12.8.3.13 The value of K_c was obtained from stability analysis of the slopes to achieve a minimum of pseudo-static FOS as detailed in **Table 12.18** and corresponding to different CTL category of the slopes which is in line with the current GEO practice.

Table 12.18 Summary of Adopted Pseudo-Static Factor of Safety (FOS)

CTL Category	Adopted-Pseudo Static FOS
Cat 1	1.1
Cat 2	1.0
Cat 3	1.0

12.8.3.14 The value of Ka is different for different S/H and frequency of vibration, where S is the velocity of travelling wave and H is the height of slope.

12.8.3.15 GEO Report No.15 [11] provided a graphical presentation of the Critical Peak Particle Velocity, PPVc, and the initial static factor of safety for varying joint displacements at peak stress, which is shown in **Diagram 12-10** below.



Notes:
Ø'p = Peak angle of shearing resistance of rock joint
(1) Corresponds to Joint Roughness Coefficient (JRC) = 5 and L=14.0m
(2) Corresponds to Joint Roughness Coefficient (JRC) = 5 and L=5.0m
(3) Corresponds to Joint Roughness Coefficient (JRC) = 5 and L=1.8m

Diagram 12-10 Plot of Critical Peak Particle Velocity against Initial Static Factor of Safety

12.8.3.16 The PPVc values adopted in this QRA are summarized as follows:

i) for slopes with CTL category 3: 25 mm/s [26];

ii) for slopes with CTL category 1 & 2:

- meet current standards: 25 mm/s [26];
- without record of upgrading works (means not sure if it meets current standards or not):
  - cut slopes that have no records of upgrading works are assigned a vibration limit of 15 mm/s.
  - fill slopes and retaining walls that have no records of upgrading works are assigned a vibration limit of 10 mm/s.

12.8.3.17 In order to calculate the slope movement due to ground vibration, a formula based on Sarma 1975 as referred in GEO Report 58 [25]. The formula for slope movement is as follow:

$$Xm = 0.25 * C * Am * T^2 * 10^{(1.07-3.83Ac/Am)}$$

Where

Xm = slope movement;
C = function of the slope geometry and generally is a value near unity;
Am = peak acceleration;
T = dominant period of the ground motion; and
Ac = critical acceleration required to cause sliding.

12.8.3.18 For blast observations, the dominant period (T) is about 1/30 seconds with peak ground acceleration in mm/s^2 is about 670 times the PPV in mm/s. It means that the peak acceleration for a PPV of 60 mm/s is about 4g or 40,000mm/s^2. Therefore, the above formula can be rewritten as follow:

$$Xm = 0.186 * PPV * 10^{(1.07-3.83PPVc/PPV)}$$

12.8.3.19 However, the formula is derived based on earthquake data, which comprised several low frequency pulses instead of a singular high frequency pulse resulted from explosives detonating. A factor of 0.25 is adopted in order to incorporate the Sarma formula in calculating slope movement due to explosives detonation as typical earthquake consists of at least four separate peaks while explosives movement mainly contains one. Therefore, the modified Sarma equation is as below:

$$Xm = 0.0465 * PPV * 10^{(1.07-3.83PPVc/PPV)}$$

12.8.3.20 The criteria for the failure of slopes based on the amount of shear displacement or slope movement based on expert judgement are summarised as follows:

- 20mm shear displacement or slope movement = 0.01% chance of slope failure
- 50mm shear displacement or slope movement = 10% chance of slope failure
- 100mm shear displacement or slope movement = 50% chance of slope failure
- 200mm shear displacement or slope movement = 100% chance of slope failure

12.8.3.21 For an estimated PPV value, the amount of slope movement can be calculated for a given slope, and hence the probability of its failure to be estimated. To be conservative, a 0.01% chance of slope failure was adopted to all slopes that would receive a shear displacement of less than 20mm.

Effect on Natural Terrains and Boulders

12.8.3.22 During blasting, the induced ground vibration may trigger boulders fall or natural terrain landslide. The Critical Peak Particle Velocity (PPVc) of a boulder will be calculated to estimate the limit of PPV that a boulder can withstand without falling.

12.8.3.23 The PPVc of boulders is calculated based on the Energy Approach with the principle of conservation of energy as shown in GEO Report No.15. The PPVc of a rock block is calculated using the following equation:

$$PPVc = (g/0.91\ (\delta p)\ \sin\beta\ (Fs/2 + 1/(2Fs) - 1)\ )^{1/2}$$

Where

g = acceleration due to gravity = 9.81 m/s^2
δp = Joint displacement at peak stress
β = Joint dip angle
Fs = initial static factor of safety

12.8.3.24 Since rock boulders can exist in various locations of the natural terrain, a sensitivity analysis approach is adopted to calculate the PPV limit of boulders that may be resting on the natural terrain. Conservative rock parameters and critical angle of natural terrain are assumed in the analysis.

12.8.3.25 Rock boulders ranging from 500 mm to 5 m in size are assessed for their critical vibration level to initiate movement. It is found that the smaller boulders result in a lower PPV limit, and this has been adopted for further analysis. A global factor of safety of 2 is then applied to the calculated vibration limit, to assign the allowable PPV of rock boulder, based on the observed natural terrain slope angle.

12.8.3.26 Boulder survey will be carried out, and assessment of specific boulder hazard will be undertaken, for all areas of natural terrain within the 5 mm/s vibration contour zone. For any boulders resting on slope greater than 30˚, the risk of instability will be individually assessed, for those areas where existing boulder survey is available. Those boulders identified as having potential instability, and also on a slope greater than 30˚, will be stabilised or protective measures will be installed, prior to the commencement of blasting.

12.8.3.27 Boulders are assumed to have 1% chance to fall when it experiences a ground vibration greater than 90 mm/s.

Effect on High Pressure Underground Town Gas Transmission Pipelines

12.8.3.28 As mentioned in **Section 12.5.5.5**, there is an interfacing section of around 9.5km between the HP pipeline and the explosives transport route (i.e. from magazine in Tai Shu Ha to Explosives Unloading Point III on Kwu Tung Road.

12.8.3.29 Leakage or rupture of a gas pipeline can be induced by a higher than expected ground vibration from an accidental explosion or during the blasting process. The typical maximum allowable PPV for town gas pipelines is 25mm/s PPV and it is considered that this level of PPV would not cause significant damage to the pipeline.

12.8.3.30 Although around 9.5km of the pipeline is under the transport route of the explosives, it is located below ground so there is no hazardous from thermal or air blast pressure effects. Moreover, there would be no shockwave transmitted into the ground as the accidental detonation would occur above ground. Therefore, the gas transmission pipeline would be able to hold up a ground vibration up to 25mm/s safely without causing any hazards.

12.8.4 Tertiary Hazards

Landslide Consequence

12.8.4.1 GEO Report No.81 Slope Failures along BRIL Roads: Quantitative Risk Assessment and Ranking [12] was published in 1999 to discuss a landslide consequence classification system. An equation was derived for the estimation of the number of fatalities:

$$N = \frac{\sum WFPEA}{V}$$

Where

W is the width of the landslide plus an adjustment for effective stopping distance;
F is the frequency of passing passengers, which may be taken as the product of the Annual Average Daily Traffic (AADT) and the average number of people in a vehicle;
P is the probability of death due to being caught in the landslide;
E is the extent of the landslide equivalent to the number of lanes affected;
A is an adjustment factor for proportion of normal road usage at the time of the landslide; and
V is the speed of vehicles.

12.8.4.2 The following assumptions were made for the above equation:

- Average vehicle speed is assumed to be 30 miles/hr (i.e. 48 km/hr) based on the road conditions in vicinity to the project site. The vehicle speed is not particularly sensitive to the calculation of N since the effect will be largely compensated by the effective stopping distance;

- A stopping distance of 23 m is assumed based on UK Highway Code data for a vehicle speed of 30 miles/hr (i.e. 48 km/hr) [13]. This stopping distance already included the reaction time;

- The probability of death due to landslides is given in **Table 12.19** below. The consequence model was developed by GEO and papers were published. Past incidents show that the assumptions are reasonable. This model was applied to several studies on landslides in Hong Kong.

Table 12.19 Probability of Fatality due to Landslide

Proximity of Slope	Probability of Death
Lane nearest to slope	0.8
2nd lane away from slope	0.6
3rd lane away from slope	0.4

- For failure of retaining wall that causes the collapse of a road, the probability of death is assumed to be 1 for the lanes affected.

- Parameter A can be 0.82 to account for the fact that landslides are most likely to occur during heavy rainfall. However, as the possible slope failure is caused by detonation of explosives in this project, the value of A is assumed to be 1.

- An adjustment factor should be applied to the calculation to account for the additional risk due to footpath adjacent to the road. It was recommended in the GEO Report No.81 that the value of N should be increased by 25% [12]. In case the footpaths being considered are comparatively remote, a lower factor than 25% is considered more appropriate. The SCL EIA [3] increased the calculated N value by 10% to account for the pedestrians on relatively remote footpaths.

12.8.4.3 Mechanism of slope failure will affect the travel distance of landslide debris. For example, a landslide induced by rainfall is expected to travel further than one caused by blasting as the soil and rock behave in a more liquid manner. Therefore, the travel distance for rainfall induced landslides may be based on an apparent angle of friction of 15˚ to 30˚. The apparent angle of friction or travel angle is defined as the inclination. The GEO Report No.81 [12] states that the travel angle of a typical rain induced landslide involves a landslide volume less than 2000 m^3 generally ranges from 30˚ to 40˚. It is conservatively assumed that a landslide caused by detonation of explosives will result in a travel angle of 30˚. The relationship of shadow / travel angle and run out distance is shown in **Diagram 12-11**.

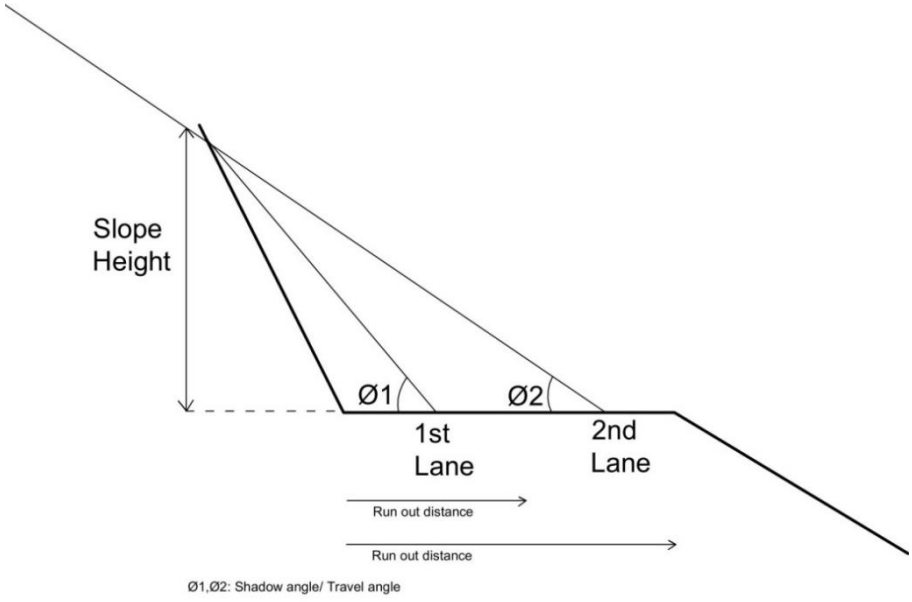


Diagram 12-11 Influence Zone for Slope Failures

12.8.4.4 Assuming the slope is a triangular volume, the run out distance for the landslide can be estimated by the following equation:

$$L = \sqrt{\frac{2V}{W\,tan(30°)}}$$

Where

L is the run out distance in m;
V is the slip volume in m^3; and
W is the slip width in m.

Boulder Fall Consequence

12.8.4.5 The estimation of boulder fall consequence is based on the methodology introduced in the GEO Report No.81 [12]. The probability of a falling rock of greater than 150mm diameter that hit a moving vehicle can be presented by the fraction of the road occupied by the vehicle, which is defined as the probability of spatial impact given a rock fall.

12.8.4.6 The probability of a moving vehicle hit by a falling rock with a diameter greater than 150mm is based on fraction of the road occupied by the vehicle which is defined as:

P(S:H) = (AADT x Length of the vehicle) / (average vehicle speed x 24,000)

Where

AADT is the annual average daily traffic (i.e. the number of vehicles per day);
Length of the vehicle is assumed to be 5m;
Average vehicle speed is assumed to be 30 miles/hr (i.e. 48 km/hr); and
Conversion factor for units is 24,000.

12.8.4.7 The above equation is modified to calculate the probability of a falling rock of greater than 150mm diameter that hit a pedestrian as below:

P(S:H) = (Number of pedestrians per day x Width of a person)/ (average walking speed x 24,000)

Where

Number of pedestrians per day is obtained by site survey;
Width of a person is assumed to be 1 m;
Average walking speed is assumed to be 5 km/hr; and
Conversion factor for units is 24,000.

12.8.4.8 The probability that a rock hits a vehicle is then given by:

$$P(S) = 1-\{1-P(S:H)\}^{N_{rf}}$$

Where

N_{rf} is the frequency of rock fall per year.

12.8.4.9 The probability of loss of life of an occupant given a vehicle is hit by a rock is assumed as 0.2 [12]. Size of the rock, number of occupants inside the vehicle and the protection of the vehicle are already considered in this probability.

12.8.4.10 Similarly, the consequence of a vehicle hitting a falling boulder can also be estimated based on the stopping distance of the vehicle. The value of stopping distance can then be substituted for that of length of vehicle in P(S:H) equation. The probability of death of an occupant is assumed as 0.1 after collision [12].

12.8.4.11 With reference to WIL study [1], it was suggested that the fatality of pedestrians hit by falling boulders is 100%.

12.8.4.12 For buildings that are located near potential boulders, the affected populations will be calculated by the proportion of the area of a boulder to the floor area of the buildings as shown in the following equation:

$$\boldsymbol{Affected\ population} = \frac{\boldsymbol{Area\ of\ a\ boulder}}{\boldsymbol{Floor\ Area\ of\ the\ building}} \times \boldsymbol{Population\ per\ floor}$$

12.8.4.13 The fatality of an occupant given a building is hit by a rock is assumed to be 20%. This figure is conservatively assumed based on the probability of loss of life of an occupant given a vehicle is hit by a rock from GEO Report No. 81 [12] as concrete walls of buildings give better protection than a vehicle.

12.8.5 Results of Consequence Assessment

Blast Effect due to Detonation of Explosives during Storage and Transport of Explosives

12.8.5.1 Consequence results of blast effects for storage and transport of explosives are summarised in **Table 12.20**. The consequence results in both Base Case and Worst Case are the same since the same amounts of explosives are stored and transported each time in both cases.

Table 12.20 Consequence Results of Blast Effect due to Accidental Detonation of Explosives during Storage and Transport of Explosives

Scenario	TNT eqv. kg	Fatality Probability	Impact Distance, m	
			Indoor	Outdoor
Storage of Explosives				
Detonation of full load of explosives in one store in Tai Shu Ha site	400	90%	23.1	18.5
		50%	26.8	19.3
		10%	39.7	21.3
		3%	53.4	22.9
		1%	68.8	24.3
Transport of Explosives				
Detonation of 200kg of explosives on one contractor truck from Magazine to explosives unloading points/ tunnel portals/ blast faces	200	90%	18.4	14.7
		50%	21.2	15.3
		10%	31.5	16.9
		3%	42.4	18.1
		1%	54.6	19.3
Detonation of 140kg of explosives on one contractor truck from Magazine to explosives unloading points/ tunnel portals/ blast faces	140	90%	16.3	13.1
		50%	18.9	13.6
		10%	28.0	15.0
		3%	37.6	16.1
		1%	48.5	17.1
Detonation of 130kg of explosives on one contractor truck from Magazine to explosives unloading points/ tunnel portals/ blast faces	130	90%	15.9	12.7
		50%	18.4	13.3
		10%	27.3	14.7
		3%	36.7	15.7
		1%	47.3	16.7
Detonation of 60kg of explosives on one contractor truck from Magazine to explosives unloading points/ tunnel portals/ blast faces	60	90%	12.3	9.8
		50%	14.2	10.2
		10%	21.1	11.3
		3%	28.4	12.1
		1%	36.5	12.9

Ground Vibration Effect on Buildings due to Errors in Blast Faces

12.8.5.2 The nearest building structure in respect to the proposed tunnel blasting sites is the village house in Ki Lun Tsuen, which is located close to the Explosives Unloading Point II and separated from the Tunnel E East Portal of around 100m. The corresponding PPV value to be received at this village house is only around 47mm/s for accidental explosion of up to 5MIC at Tunnel E. This predicted PPV value is much lower than the building structural element collapse threshold (i.e. 229 mm/s) and the falling object threshold (i.e. 100 mm/s). No major damage is thus expected and the impact to building structure was not further assessed in this QRA.

Ground Vibration Effect on Slopes due to Errors in Blast Faces

12.8.5.3 The corresponding influence zone of the screening criterion of PPV=90 mms/s for slopes due to error in tunnel blasting are presented in **Table 12.21**.

Table 12.21 Influence Zone of 90 mm/s from 2 MIC to 5 MIC for Effect on Slopes

PPV (mm/s)	Influence Zone (m)			
	At Blast Face			
	2 MIC (20kg)	3 MIC (30kg)	4 MIC (40kg)	5 MIC (50kg)
90	37.4	45.8	52.9	59.1

12.8.5.4 The features within the influence zone of 90 mm/s of all blasting sites were reviewed. Some surround slope features were identified to receive a PPV level that cause potential failure during the blasting activities at Tunnel E, and the affected slopes are summarised in **Table 12.22**.

Table 12.22 Slopes affect by Higher than Expected Vibrations during Construction of Tunnel E

Scenario / Slope No.	Slope Type	Height (m)	Length (m)	Angle (deg)
3 MIC detonated at the same time				
2SE-D/C 145	Cut Slope	6	7	45
2SE-D/C 146	Cut Slope	4	5	60
4 MIC detonated at the same time				
2SE-D/C 140	Cut Slope	13	20	50
2SE-D/C 145	Cut Slope	6	7	45
2SE-D/C 146	Cut Slope	4	5	60
5 MIC detonated at the same time				
2SE-D/C 140	Cut Slope	13	20	50
2SE-D/C 145	Cut Slope	6	7	45
2SE-D/C 146	Cut Slope	4	5	60

Ground Vibration Effect on Boulders due to Errors in Blast Faces

12.8.5.5 The proposed tunnels are to be constructed under some natural terrains. The corresponding influence zones of 90 mm/s due to error in tunnel blasting can refer to **Table 12.21**. An assessment area of 100m was adopted and is presented in

Appendix 12.3. This 100m assessment area well covered the influence zone of 90 mm/s due to 5 MIC in tunnel blasting.

12.8.5.6 The shortest separation distance between natural terrains and blast faces is less than 10m, boulder fall potentially exists for the errors of blast faces during construction of the tunnels.

12.9 Risk Evaluation

12.9.1 Overview

12.9.1.1 The risk summation considered the frequency and consequence assessment including the risk to nearby building and road populations, and the results are presented in individual risk contours, FN curves and Potential Loss of Life (PLL).

12.9.1.2 Individual risk is a measure of the risk to a chosen individual at a particular location. As such, this is evaluated by summing the contributions to that risk across a spectrum of incidents which could occur at a particular location.

12.9.1.3 Societal risk is a measure of the overall impact of an activity upon the surrounding community. As such, the likelihoods and consequences of the range of incidents postulated for that particular activity are combined to create a cumulative picture of the spectrum of the possible consequences and their frequencies. This is usually presented as an FN curve and the acceptability of the results can be judged against the societal risk criterion under the risk guidelines.

12.9.2 Risk Results

Blast and Overpressure Waves Generated due to Accidental Detonation during Storage at Magazine Site and Transport of Explosives to Blast Faces

12.9.2.1 The societal risk data of blast effect during storage and transport of explosives are presented in **Table 12.23**.

Table 12.23 Societal Risk Data of Blast Effect during Storage and Transport of Explosives

Number of Fatalities	Frequency with N or More Fatalities (per year)	
	Storage of Explosives	Transport of Explosives
1	2.54E-07	1.25E-05
2		9.86E-06
3		7.24E-06
4		5.62E-06
5		4.45E-06
6		3.31E-06
8		1.66E-06
10		1.08E-06
12		7.62E-07
15		4.49E-07
20		1.57E-07
25		5.44E-08
30		1.51E-08
40		1.44E-08
50		1.21E-08
60		1.21E-08
80		8.36E-09
87		5.73E-10

Ground Vibration Effect on Slope due to Errors in Blast Faces

12.9.2.2 The results of scenario frequencies and expected fatalities for the affected slopes due to ground shock generated because of errors in the blast faces are summarised in **Table 12.24**. The calculated slope movement ranged from 6.3mm to 16.7mm.

Table 12.24 Slopes affect by Higher than Expected Vibrations during Construction of Tunnel E

Scenario / Slope No.	Max. PPV received (mm/s)	Xm (mm)	Chance of Slope Failure (%)	Run Out Distance (m)	Slope Failure Frequency (per year)	Expected Fatality (N) [1]
3 MIC detonated at the same time						
2SE-D/C 145	104.7	7.0	0.01	7.9	3.68E-11	0
2SE-D/C 146	101.5	6.3	0.01	4.0	3.68E-11	0
4 MIC detonated at the same time						
2SE-D/C 140	107.1	7.5	0.01	15.7	7.36E-13	0
2SE-D/C 145	124.8	11.6	0.01	7.9	7.36E-13	0
2SE-D/C 146	121.0	10.7	0.01	4.0	7.36E-13	0
5 MIC detonated at the same time						
2SE-D/C 140	122.7	11.1	0.01	15.7	7.37E-15	0
2SE-D/C 145	143.0	16.7	0.01	7.9	1.84E-14	0
2SE-D/C 146	138.7	15.4	0.01	4.0	1.47E-14	0

Note:

[1] Slopes 2SE-D/C 140, 2SE-D/C 145, 2SE-D/C 146 and 2SE-D/C 148 are located at a distance of 44m to 90m away from the nearest off-site population, i.e. the construction workers in the San Tin Technopole, slope run out from these slopes will not reach any off-site population.

Ground Vibration Effect on Boulders due to Errors in Blast Faces

12.9.2.3 The calculated frequencies for boulder fall due to errors in blast faces are summarised in **Table 12.25** below.

Table 12.25 Boulder Fall Frequencies for Accidental Initiation of Explosives from 2 MIC to 5 MIC due to Errors in Blast Faces

Works Area	2 MIC	3 MIC	4 MIC	5 MIC
Tunnel A	8.32E-06	1.18E-07	1.66E-09	2.11E-11
Tunnel B	3.60E-05	4.32E-07	5.25E-09	5.53E-11
Tunnel C	1.32E-06	6.72E-08	9.51E-10	9.51E-12
Tunnel E	6.96E-06	8.47E-08	1.14E-09	1.14E-11

12.9.2.4 With consideration of the topography, mis-blast during construction of the tunnels will result in boulder fall to areas close to the blasting sites, including Ngau Tam Mei, Ki Lun Tsuen, Cheung Lek, Kwu Tung and San Tin areas, and are summarized in **Table 12.26** to **Table 12.30**.

Table 12.26 Occurrence Frequencies for a Boulder Falling to Ngau Tam Mei Area

Works Area / Offsite Risk Receiver	2 MIC	3 MIC	4 MIC	5 MIC
Tunnel A				
Catchments A2 [1]	8.32E-06	9.28E-08	1.15E-09	1.28E-11

Notes:

[1] With consideration of the topography, any accidental initiation of explosives from 2MIC to 5MIC for blasting of Tunnel A may result in boulder fall into the catchments A2 (refers to **Appendix 12.8**), which channelises to the flat land near the northeastern boundary of the Ngau Tam Mei Development site. Since this area will be the works area of this Project, no off-site individual is expected in the potentially affected area.

Table 12.27 Occurrence Frequencies for a Boulder Falling to Ki Lun Tsuen Area

Works Area / Offsite Risk Receiver	2 MIC	3 MIC	4 MIC	5 MIC
Tunnel A				
Catchments A14 / A15	0	9.60E-09	2.24E-10	3.52E-12
Tunnel B				
Catchment B3	1.68E-06	4.96E-08	7.09E-10	8.51E-12
Catchment B7 [1]	2.68E-05	3.05E-07	3.61E-09	3.76E-11
Catchment B9 [1]	7.54E-06	7.80E-08	9.21E-10	9.21E-12
Tunnel C				
Catchment B9 [1]	1.32E-06	3.36E-08	4.48E-10	4.48E-12
Tunnel E				
Catchments E2 / E5 [2]	6.96E-06	7.37E-08	8.10E-10	8.10E-12
Catchment E6	0	0	7.36E-11	7.37E-13

Notes:

[1] With consideration of the topography, any accidental initiation of explosives from 2MIC to 5MIC for blasting of Tunnel B or Tunnel C which result in a potential boulder fall into the catchments B7/B9 (refers to **Appendix 12.8**), which channelise to the Kwu Tung Reservoir with no off-site individual expected in the potentially affected area.

[2] With consideration of the topography, any accidental initiation of explosives from 2MIC to 5MIC for blasting of Tunnel E may result in boulder fall into the catchments E2/E5 (refers to **Appendix 12.8**), which channelises to the flat land near Explosives Unloading Point II. Since this area will be the works area of this Project, no off-site individual is expected in the potentially affected area.

Table 12.28 Occurrence Frequencies for a Boulder Falling to Cheung Lek Area

Works Area / Offsite Risk Receiver	2 MIC	3 MIC	4 MIC	5 MIC
Tunnel A				
Catchment A11	0	9.60E-09	2.88E-10	4.80E-12

Table 12.29 Occurrence Frequencies for a Boulder Falling to Kwu Tung Area

Works Area / Offsite Risk Receiver	2 MIC	3 MIC	4 MIC	5 MIC
Tunnel C				
Catchment C4	0	3.36E-08	5.04E-10	5.04E-12

Table 12.30 Occurrence Frequencies for a Boulder Falling to San Tin Area

Works Area / Offsite Risk Receiver	2 MIC	3 MIC	4 MIC	5 MIC
Tunnel E				
Catchments E4 / E9 [1]	0	1.10E-08	2.58E-10	2.58E-12

Notes:
[1] Site formation works under the Project will be carried out for the concerned natural terrain area, potential for boulder fall would be eliminated.

12.9.2.5 The result of scenario frequencies and expected fatalities for boulder fall due to errors in blasting faces during construction of the 4 tunnels are summarised in **Table 12.31**. Considering a boulder with size of 150mm diameter, a fatality is assumed for a boulder hitting a building or a pedestrian. The off-site population affected are shown in **Appendix 12.8**. The overall frequency of 1 fatality due to boulder fall is 2.16×10^{-7} per year.

Table 12.31 Scenario Frequencies and Expected Fatalities for Boulder Fall due to Errors in Blast Faces

Scenario / Road & Building	Scenario Frequency (per year)	Expected Fatality (N)
Detonation of 2 MIC to 5 MIC at Tunnel A		
Village House in Ki Lun Tsuen	1.97E-09	1
Camping Site in Cheung Lek	9.90E-09	1
Detonation of 2 MIC to 5 MIC at Tunnel B		
Village House in Ki Lun Tsuen	3.45E-07	1
Detonation of 2 MIC to 5 MIC at Tunnel C		
Village House in Kwu Tung	6.82E-09	1
Detonation of 2 MIC to 5 MIC at Tunnel E		
Ki Lun Tsuen Playground	7.44E-11	1

12.10 Risk Summation

12.10.1 Individual Risk

12.10.1.1 The individual risk (IR) contours for the storage, transport and use of explosives are shown in **Appendix 12.7**. The IR contours shown are for indoor population since the impact zones for indoor population are larger than that for outdoor population. The maximum individual risk along the transport routes is 1×10^{-8} per year while that the maximum individual risk of 1×10^{-5} per year at the magazine site. Although the 1×10^{-5} per year contour associated with the overnight storage of explosives at the magazine site is beyond the boundary of the magazine, the impact area is only on woodland areas where there is no continuous presence of people. As a result, no off-site individual will be exposed to an individual risk of more than 1×10^{-5} per year. On this basis, the level of individual risk associated with the storage, transport and use of explosives should be acceptable since it meets the HKRG.

12.10.2 Societal Risk

12.10.2.1 The FN curves for the storage, transport and use of explosives, as well as the overall risk are shown in **Diagram 12-12**, while their associated data points are summarised in **Table 12.32**. The FN curves for the Worst Case scenario are shown in **Diagram 12-13** and the societal data points are tabulated in **Table 12.33**. It can be seen that the risk due to transport of explosives lies in ALARP region, while the risk associated with storage and use of explosives lies in Acceptable region.

Table 12.32 Societal Data Points of FN Curve for Storage, Transport and Use of Explosives (Base Case Scenario)

No. of fatalities	Frequency of accidents with N or more fatalities per year					
	Blast Effect due to Storage of Explosives	Blast Effect due to Transport of Explosives	Slope Failure	Boulder Fall	Falling Objects	Cumulative
1	2.54E-07	1.28E-05	0	3.64E-07	0	1.34E-05
2		1.00E-05				1.00E-05
3		7.39E-06				7.39E-06
4		5.71E-06				5.71E-06
5		4.52E-06				4.52E-06
6		3.38E-06				3.38E-06
8		1.74E-06				1.74E-06
10		1.17E-06				1.17E-06
12		8.49E-07				8.49E-07
15		5.36E-07				5.36E-07
20		2.36E-07				2.36E-07
25		1.30E-07				1.30E-07
30		7.21E-08				7.21E-08
40		4.43E-08				4.43E-08
50		2.05E-08				2.05E-08
60		1.69E-08				1.69E-08
80		8.36E-09				8.36E-09
87		5.73E-10				5.73E-10

Table 12.33 Societal Data Points of FN Curve for Storage, Transport and Use of Explosives (Worst Case Scenario)

No. of fatalities	Frequency of accidents with N or more fatalities per year					
	Blast Effect due to Storage of Explosives	Blast Effect due to Transport of Explosives	Slope Failure	Boulder Fall	Falling Objects	Cumulative
1	2.54E-07	2.56E-05	0	7.28E-07	0	2.66E-05
2		2.01E-05				2.01E-05
3		1.48E-05				1.48E-05
4		1.14E-05				1.14E-05
5		9.04E-06				9.04E-06
6		6.75E-06				6.75E-06
8		3.48E-06				3.48E-06
10		2.33E-06				2.33E-06
12		1.70E-06				1.70E-06
15		1.07E-06				1.07E-06
20		4.71E-07				4.71E-07
25		2.60E-07				2.60E-07
30		1.44E-07				1.44E-07
40		8.86E-08				8.86E-08
50		4.10E-08				4.10E-08
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80		1.67E-08				1.67E-08
87		1.14E-09				1.14E-09

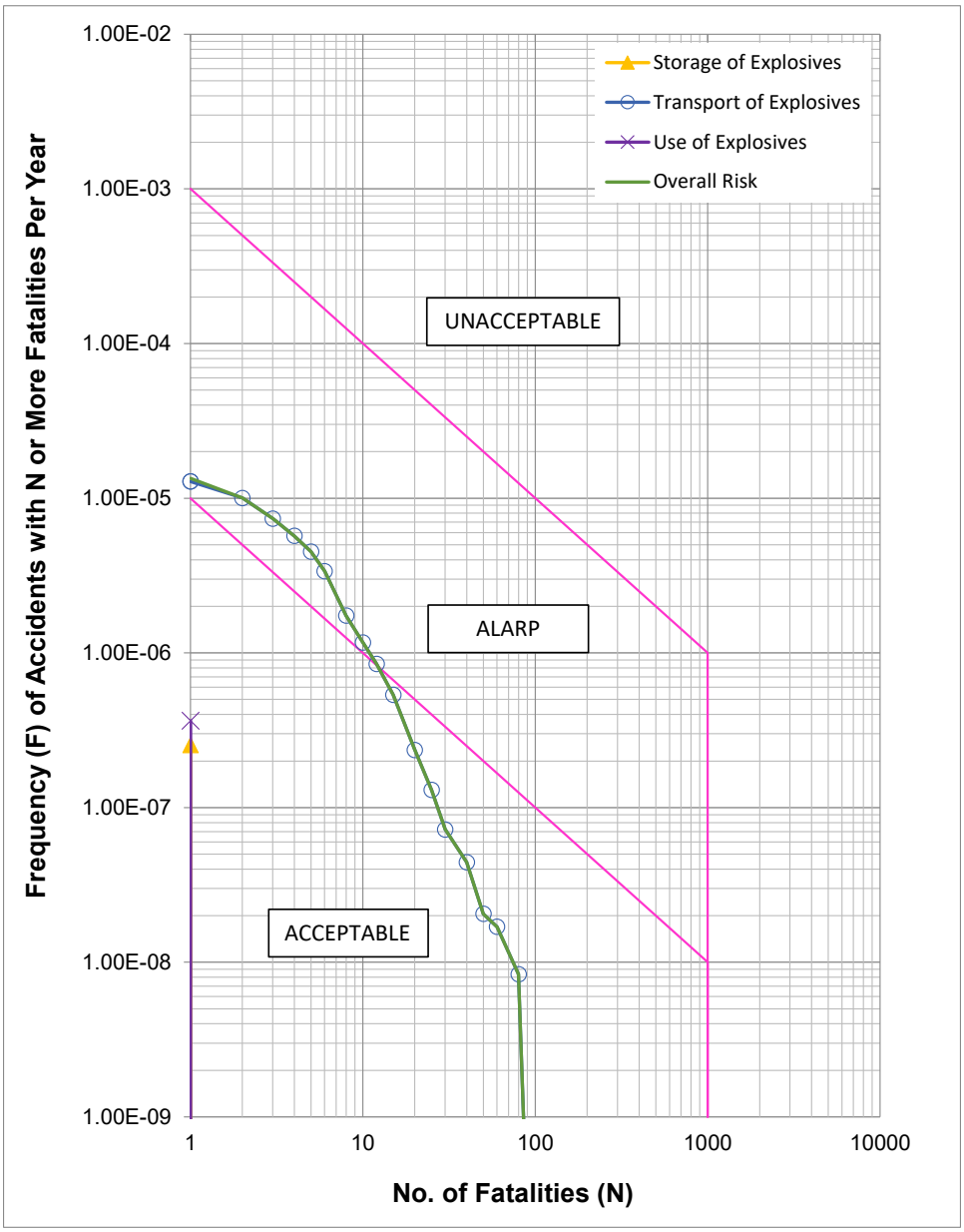


Diagram 12-12 FN Curve for Storage, Transport and Use of Explosives (Base Case Scenario)

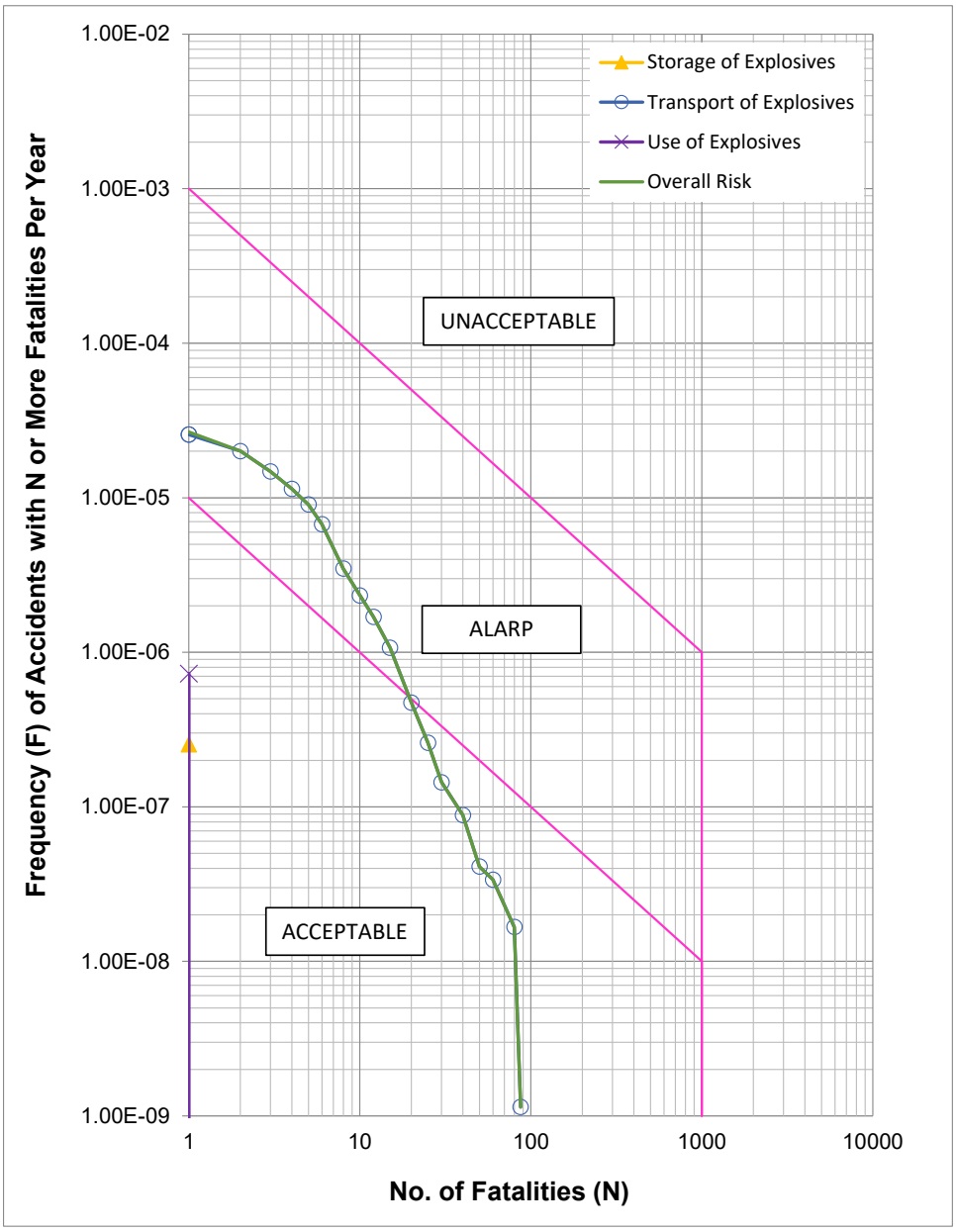


Diagram 12-13 FN Curve for Storage, Transport and Use of Explosives (Worst Case Scenario)

12.10.2.2 The potential Loss of Life (PLL) value is the summation of the product of each FN pair. The breakdowns of PLL are presented in **Table 12.34**. The Worst Case scenario with consideration of 100% increase of the number of delivery trips and also the number of blasts is also presented in the table. Transport of explosives is the major contributor of risk and it contributes to over 99% of the overall risk in both Base Case scenario and Worst Case scenario.

Table 12.34 Breakdown of PLL

	Base Case		Worst Case	
Scenario	**PLL Contribution**	**PLL %**	**PLL Contribution**	**PLL %**
Storage of Explosives	2.54E-07	0.4	2.54E-07	0.2
Transport of Explosives	5.84E-05	99.0	1.17E-04	99.2
Use of Explosives	3.64E-07	0.6	7.28E-07	0.6
Total	**5.90E-05**	**100**	**1.18E-04**	**100**

12.11 Risk Mitigation Measures

12.11.1 Risk Results

12.11.1.1 The hazard to life assessment of the Project assessed the risks arising from the storage, transport and use of explosives. From Section 12.10, the risks posed by the transport of explosives falls within the lower "ALARP" region specified in EIAO-TM Annex 4, risk mitigation measures should be investigated.

12.11.2 Approach

12.11.2.1 Practicability of risk mitigation measures is usually evaluated by cost-benefit analysis, which is a trade-off between the risk mitigation and the cost of the risk mitigation measure.

12.11.2.2 In this Project, CBA, if needed, would be applied by calculating the implied cost of averting a fatality (ICAF) for various mitigation measures identified. The ICAF value is calculated as follows:

$$ICAF = \frac{Cost\ of\ mitigation\ measure}{Reduction\ in\ PLL\ value \times Design\ life\ of\ mitigation\ measure}$$

12.11.2.3 ICAF is a measure of the cost per life saved over the lifetime of the project due to implementation of a particular mitigation measure. It may be compared with the value of life to determine whether a mitigation measure is reasonably practicable to implement, i.e. if ICAF is less than the value of life, then the mitigation measure should be implemented. In this study the value of life is taken as HK$33M, which is the same figure as used in previous risk assessment reports including QRA of STC STW EIA. An aversion factor is taken as 20 and the adjusted value of life using this aversion factor is HK$660M. This is the value to measure how much the society is willing to invest to prevent a fatality, where there is potential for an event to cause multiple fatalities.

12.11.2.4 To identify whether the mitigation measures, the Maximum Justifiable Expenditure (MJE) is estimated for determining whether a mitigation measure is justifiable. The value of MJE will be calculated by assuming the risk is reduced to zero while justifiable mitigation measures will be examined by considering its actual reduction in PLL in the calculation of safety benefit. The equation of MJE is as follows:

$$MJE = Valueof\ Preventing\ a\ fatality \times Aversion\ Factor \times Reduction\ in\ PLL\ value \times Design\ life\ of\ mitigation\ measure$$

12.11.2.5 The mitigation measures will be considered as justifiable if the cost is less than the value of MJE and the implementation cost for this option is not more than the calculated safety benefits.

12.11.2.6 The capital and operational costs should only be included as the implementation cost of mitigation measures. Any costs related to design or change of design should be excluded.

12.11.2.7 Some mitigation measures may not be able to quantify the cost-benefits, so a qualitative approach will be used in these cases.

Maximum Justifiable Expenditure

12.11.2.8 The MJE for this Project is calculated with a conservative aversion factor of 20.

$MJE = HK\$33M \times 20 \times 1.17\times10^{-4} \times 5$

$= HK\$0.39M$

12.11.2.9 The reduction in PLL is based on the total PLL of transport of explosives for the Worst Case, while design life is assumed to be 5 based on the construction phase of this Project during which transport of explosives will be involved.

12.11.2.10 For a mitigation measure to be potentially justifiable, its cost should be less than the Maximum Justifiable Expenditure.

12.11.3 Potential Justifiable Mitigation Measures

12.11.3.1 The approach considered the identification of options pertaining in the following broad categories:

- Options of using alternative methods of construction such as Tunnel Boring Machine (TBM);
- Options of using magazines closer to the construction sites;
- Options of using different explosive types;
- Options of reducing the quantities of explosives to be used;
- Options considering improved design of the explosives delivery vehicle; and
- Options considering better risk management systems and procedures.

Use of Alternative Methods of Construction

12.11.3.2 One alternative method of construction is adoption of hard rock tunnel boring machines for constructing hard rock tunnel. However, the largest hard rock TBM machine evert built in the world is around 15m. It cannot cater for the total span requirement of minimum 16m for the proposed 3-lane highway road for this project. It is considered not technically feasible.

12.11.3.3 Another alternative method of construction is mechanical breaking for the hard rock tunnel. However, this excavation rate of mechanical breaking is only 1-3m/day,

which is lower than the excavation rate with drill-and blast method of 5m/day. This method of construction could not cater for the construction programme of the project.

12.11.3.4 Therefore, this option is not practicable.

Use of Magazines Closer to the Construction Sites

12.11.3.5 Amongst the initial proposed list of Magazine sites (as indicated in **Appendix 12.9**), only the Tai Shu Ha (Yuen Long) site is practicable.

12.11.3.6 The transportation route to Tai Shu Ha Road West (~11km) is the closest among all other magazine sites. This would greatly reduce the risk for transportation of explosives.

12.11.3.7 Therefore, Tai Shu Ha (Yuen Long) site is recommended.

Use of Different Explosives Types

12.11.3.8 The type of explosives used in the Project is already considered as the safest type for blasting application, so no safety benefits will be obtained by selecting a different type of explosives.

12.11.3.9 In this project, PETN is used as the detonating cord with a melting point around 140°C. Although there are other detonating cord technologies available such as Research Department explosive (RDX) and High Melting Explosive (HMX), their melting points are slightly higher than that of PETN which may require more time before an explosion occurs following a fire event. However, the extra time required and risk reduction by implementing this mitigation measure would be neglected for the purpose of this assessment.

12.11.3.10 Hence, this option is not considered further.

Use of Smaller Quantities of Explosives

12.11.3.11 This project has already minimized the quantity of explosives required for transportation. Only initiating explosives will be transported, while bulk emulsion explosives will be manufactured on site. An alternative option is to use cast boosters as primers in place of cartridged emulsion for initiating bulk emulsion blasting. Adoption of cast boosters would reduce the amount of explosive material transported per delivery.

12.11.3.12 The primary energetic component of cast boosters is PETN, which has a higher TNT equivalency than cartridged emulsions. Furthermore, the use of cast boosters would not remove the need for detonating cord within the blasting system.

12.11.3.13 Based on information provided by the supplier, the unit cost of a cast booster is approximately HK$7.5 higher than that of cartridged emulsion. Given that more than 370,000 cast boosters would be required for this project, the additional expenditure is estimated to exceed HK$2.8 million compared with the use of cartridged emulsion as the primer.

12.11.3.14 This additional cost significantly exceeds the Maximum Justifiable Expenditure and therefore cannot be supported on economic grounds. However, the use of cartridge

emulsion or booster as the initiative explosives is subject to the contractor's explosive selection.

Safer Design of the Explosives Delivery Vehicles

12.11.3.15 The design of the explosives delivery vehicles has been reviewed to identify potential improvements which can reduce the risk such as preventing the fire spreading to the explosive load. It is assumed that the explosives delivery vehicle has already followed the guideline provided by Mines. For example, an additional vertical fire screen with heat resistance equivalent to 3mm of steel should be installed between the vehicle cabin and the cargo compartment of contractor's truck.

12.11.3.16 Besides assuring the contractor's truck has fully complied the requirement of Guidance Note No. GN2 published by Mines. There are also some simple measures can be implemented. For instance, limiting the fuel tank capacity and reducing the combustible load on the explosives by using fire retardant materials wherever possible.

12.11.3.17 Since the safety benefits of these measures are difficult to evaluate quantitatively, they are included in the section regarding good practices (**Section 12.11.5**).

Better risk management systems and procedures

12.11.3.18 The accident involvement frequency of the explosives delivery vehicle can be reduced through the training programme for both the driver and his attendants, regular "toolbox" briefing sessions, implementation of a defensive driving attitude, appropriate driver selection based on good safety record and regular medical checks.

12.11.3.19 The fire involvement frequency can be reduced by putting better types of fire extinguishers with bigger capacity inside the explosive delivery vehicles.

12.11.3.20 Moreover, it can be reduced by providing emergency plans and training to ensure that the suitable fire extinguishers are used and attempt is made to evacuate the area of the incident or securing the explosive load if possible.

12.11.3.21 The actual implementation of this option is provided in the recommendation section (**Section 12.11.5**).

Summary

12.11.3.22 In summary, most of the options considered for CBA are not practicable after comparing the implementation cost with the MJE, while the remaining options have been recommended as Good Practices recommended under **Section 12.11.5**.

12.11.4 ALARP Assessment Results

12.11.4.1 The evaluation of each option considered is concluded in **Table 12.35**.

Table 12.35 ALARP Assessment

Mitigation Measures	Practicability	Implementation Cost	ALARP Assessment Results
Use of alternative method of construction	Not practicable	-	TBM is considered technically not feasible and mechanical breaking excavation rate is much slower than drill-and-blast method.
Use of magazines closer to the construction sites	Not practicable	-	Closest practicable magazine site to the construction sites has been selected.
Use of different explosives types	Pose some limitations	-	Bulk emulsion explosives manufactured on site has been adopted as the main or `bulk' blasting explosive to excavate rock by rock blasting to minimize the amount of explosives for transportation.
Use of smaller quantities of explosives	Not practicable	>HK$ 2.8M	Use of cast boosters is not cost effective. However, due to the risk reduction achieved the use of cast boosters is to be maximized, which is to be encouraged. In the other hand, Project Proponent, i.e. Highways Department, would review, on an ongoing basis during the detailed design, tender and construction phases, and implement the use of cast boosters during construction to the maximum extent possible.
Safer design of the explosives delivery vehicles	Practicable	-	Based on low implementation costs, this mitigation option has been directly incorporated in recommendations (under **Section 12.11.5**)
Better risk management systems and procedures	Practicable	-	Based on low implementation costs, this mitigation option has been directly incorporated in recommendations (under **Section 12.11.5**)

12.11.5 Recommendations

Recommendations for Meeting the ALARP Requirements

12.11.5.1 Recommendations are made for implementation in order to meet the EIAO-TM requirements:

- The explosives delivery vehicles should be designed and improved to reduce the amounts of combustibles in the cabin. The fuel carried in the fuel tank should also be minimized to reduce the duration of any fire.

- The accident frequency of the explosive delivery vehicles should be minimized through the implementation of a defensive driving attitude and a dedicated training programme for both driver and his attendants which includes regular briefing sessions. Moreover, drivers should be selected based on good safety record and providing regular medical checks for the driver.

- The required quantity of explosives should only be transported for a particular blast to avoid any unused explosives send back to the magazine.

- The contractor should combine the explosive deliveries for a given work area as far as practicable.

- A minimum headway between two consecutive vehicle convoys of 10 mins should be maintained whenever practicable.

- To reduce the fire involvement frequency of explosive delivery vehicles, a better emergency response and training should be implemented to ensure adequate fire extinguishers are used and attempt is made to evacuate the area of the incident or securing the explosive load if possible. All explosive delivery vehicles should also be equipped with bigger capacity AFFF-type extinguishers.

General Recommendations

12.11.5.2 Each blasting activities including storage and transport of explosives should be supervised and audited by competent site staff to ensure strict compliance with the blasting permit conditions.

12.11.5.3 For the storage and transport of explosives, the recommendation listed below should also be considered:

- The security plan should address different alert security level to reduce opportunity for arson or deliberate initiation of explosives.

- Emergency plan like magazine operation manual should be developed to address uncontrolled ire in magazine area and during transport of explosives.

- Adverse weather working guideline should be developed to clearly define procedure for transport of explosives during thunderstorm.

Good Practices to be Implemented for Storage of Explosives

12.11.5.4 While the risk levels associated with the overnight storage of explosives for this Project is minimized as much as practicable that the Contractor implements all the good practices to minimize the potential risks even further and ensure that overnight store of explosives will not adversely affect services, utilities, slopes, retaining walls, buildings and structures through ground vibrations or other effects. A summary of these good practices is given below for reference.

12.11.5.5 The good practice could be made reference to the latest guideline including, but not limited to, “Guidance Note No. GN 8 How to Apply for a Mode A Store Licence for Storage of S1DG (Blasting Explosives)” by CEDD. The design, operation and maintenance of the magazine should follow Mines Division guidelines and industry best practice. Some other good practices listed below can also be implemented:

- To ensure the undertaken work activities during the operation of the magazine are properly controlled, a suitable work control system such as an operational manual including Permit-to-Work system should be introduced.

- Good housekeeping should be maintained within the magazine to ensure that combustible materials are not allowed to accumulate.

- The magazine store should not have any open drains, traps, pits or pocket which any molten ammonium nitrate could flow and be confined in the even of a fire.

- Regular checking of the magazine building should be conducted for water seepage through the roof, walls or floor.

- Caked explosives shall be disposed of in an appropriate manner.

- Permission to remain the secured fenced off magazine store area shall not be given to explosives delivery vehicles.

- Speed limit control should be implemented within the magazine area in order to reduce the risk of a vehicle impact or incident within the magazine area.

Good Practices to be Implemented for Transport of Explosives

12.11.5.6 Contractor should implement all good practices to minimize the potential risks even further and ensure that transport of explosives will not result in adverse impact. A summary of these good practices is given below for reference. The good practice could be made reference to the latest guideline including, but not limited to "Guidance Note No. GN 2 Approval of an Explosives Delivery Vehicle" and "Guidance Note No. GN 3 Application and Handling of a Conveyance Permit" by CEDD:

- Typical Removal Permit Conditions

    - A placard as specified in the section 80 of Dangerous Goods (Control) Regulation must be displayed in a conspicuous place on the vehicle carrying explosives.
    - No unnecessary waiting or parking of the vehicle is permitted at any place along the transportation route.
    - The vehicle carrying the explosives is prohibited from passing through any tunnel on a public road.
    - Except with the permission in writing of the Authority, the vehicle must not carry more than 200kg net explosives content of explosives at any one time. The vehicle for moving explosives shall be a licensed vehicle equipped with effective fire-extinguishers and maintained in good running conditions at all time.
    - The vehicle shall use the intended route of transportation specified in the application for this conveyance permit.
    - The vehicle with explosives on board is prohibited from refuelling at any fuel station.
    - Conveyance of blasting explosives or entertainment fireworks shall only be undertaken by the vehicle/s and driver/s approved by the Authority and in the presence of a Resident Explosives Supervisor and a Shot Firer or a Fireworks Master/Assistant. When carrying explosives/fireworks, the approved vehicle/s shall display the correct dangerous goods placards and warning signs.

  - Explosives and detonators must be conveyed on separate vehicles or in separate compartments on the vehicle. Electric detonators must be carried in an approved and properly labelled wooden container.
  - The Permittee is required to input the actual date and time of the use of this Permit in Centralised Explosives Licensing and Management System (CELIMS) after the conveyance of the explosives as soon as reasonably practicable. If the Permit is unused before its expiry date, the Permittee is also required to provide reason(s) for not using the Permit in CELIMS.

- Safer Design of the Explosives Delivery Vehicle
  - Fire screen could be installed between the cabin and the load of the vehicle to reduce the chance of fire escalating to the load and cause explosion.
- Reduction of Accident Involvement Frequency
  - Different administrative measures can be implemented to reduce the accident involvement frequency and increase the situational awareness of the driver during the transportation of explosives.
  - Administrative measures can include “Tool-box” talk training regarding the safety precautions when transporting explosives.
  - Ensuring that the detonators and the cartridged emulsion are under good conditions and well-intact within their packaging before transporting.
  - Recruiting experienced driver with good safety record and checking their health condition in a regular basis.
- Reduction of Fire Involvement
  - Carrying fire extinguishers or other active fire protection devices with higher standard and higher capacity onboard of the explosives delivery vehicles.
  - Create a contingency plan with consideration of different scenarios that may occur, such as the action that the driver should take in case of fire near the explosives delivery vehicles in the middle of traffic jam.
  - Regulations for the drivers should be set, such as hot work should be prohibited when handling explosives to avoid any sources of ignition.
  - Working Guidelines should be developed to provide clear instructions to the drivers when encountering different situations like extreme weather.

Good Practices to be Implemented for Use of Explosives

12.11.5.7 While the risk levels associated with the use of explosives for this Project is minimized as much as practicable, it is prudent that the Contractor implements all the good practices to minimize the potential risks even further and ensure that any blasting carried out will not adversely affect services, utilities, slopes, retaining walls, buildings and structures through ground vibrations or other effects. A summary of these good practices is given below for reference. The good practice could make reference to the latest guideline including, but not limited to, Practice Note for

Authorized Persons and Registered Structural Engineers – Control of Blasting (APP-72) by Buildings Department (BD). Following are some Typical Items regarding Good Practices to Blasting Works extracted from the APP-72, for detail, please reference to the latest APP-72 by BD.

- Carry out checking of the registered contractor's blasting method statement.
- Check (including both document and site checks) and satisfy, for each blast, that the registered contractor's blast design and preventive measures comply with the plans approved by the Building Authority and the blasting permit requirements.
- Verify on site that the ground conditions and geology are as stated or assumed in the blasting assessment, and that the provisions in the method statement and the preventive, protective and preventive measures are adequate for the conditions as encountered on site.
- Ensure that the preventive measures, if required, have been properly carried out prior to commencement of the blasting works.
- Prepare regular reports with records of the condition of the site, sensitive receivers, adjacent grounds, structures and services etc. after each phase of blasting operation and completion of related works.
- Inspect the construction of preventive works, if required, for the sensitive receivers.
- Inspect the provision and installation of all necessary protective and preventive measures prior to each blast, in accordance with the blast design.
- Monitor the site operations and working methods to ensure that they meet the safety requirements set out in the blasting permit.
- Inspect and monitor the conditions of all sensitive receivers regularly and carry out reviews of the quality of monitoring for the sensitive receivers before and after each blast.

12.12 Conclusions

12.12.1.1 A QRA was conducted to evaluate the risk induced from the storage, transport and use of explosives during construction of the Project.

12.12.1.2 To ensure timely delivery to blasting site and maintain the construction process, a temporary explosive magazine site at Tai Shu Ha (Yuen Long) for overnight storage of explosives is required.

12.12.1.3 The magazine site at Tai Shu Ha (Yuen Long) would follow the relevant design requirements in terms of sufficient separation distances and the design of the storage facilities. Review suggests that the associated risk for the operation of the temporary magazine site would be insignificant.

12.12.1.4 The assessment results show that the individual risks associated with the storage, transport and use of explosives complied with the Risk Guidelines as stated in Annex 4 of the EIAO-TM.

12.12.1.5 The societal risks associated with the storage and use of explosives fall into the "Acceptable" region of the Risk Guidelines; while the societal risk associated with the transport of explosives falls into the lower "ALARP" region of the Risk Guidelines, mitigation measures for the transport of explosives were explored and assessed by using CBA. Justified mitigation measures and good practices were recommended. No residual risk impact is anticipated.

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